Saving Responses to Mandatory Pension Plans*

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Abstract

To boost retirement savings, many countries mandate worker contributions to pension accounts. This paper investigates saving responses to such mandates throughout the entire portfolio, leveraging detailed administrative tax data from Switzerland and a regression discontinuity design. I find that mandatory pension plans have limited effects on total savings, with an estimated crowd-out rate of 73%. Decomposing the saving response, I show that workers offset mandatory pension contributions by reducing private non-retirement savings, primarily in financial assets. Liquidity-constrained workers are less able to decrease their private savings in response to mandatory contributions, implying that the mandate raises their total savings.

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1 Introduction

Population aging is putting pressure on pension systems, making it increasingly difficult to provide adequate financial support to retirees (Poterba, 2014). In response, many governments have implemented policies aimed at boosting retirement savings during individuals' working lives, including mandates that require contributions to pension accounts. As of 2022, 19 of the 38 OECD countries had adopted some form of mandatory pension plans (OECD, 2023). Even in countries without a comprehensive mandate, such as the US, some employer-sponsored pension plans feature mandatory contributions (Card and Ransom, 2011; Friedberg, Leive and Cai, 2024). Typically, these mandates require workers to contribute a fixed share of their earnings, with contributions withheld at source by employers and deposited directly into illiquid pension accounts. This facilitates the accumulation of retirement savings but reduces workers' take-home pay.

Do pension plan mandates succeed in boosting savings? Ex ante, this is unclear. Individuals may respond by reducing other types of savings or taking on additional debt to maintain the same optimal savings level as in the absence of the mandate. In such cases, mandatory contributions crowd out other savings, altering the portfolio composition without changing overall savings levels. Therefore, it is key to analyze saving responses across the entire portfolio to understand how mandatory pension plans affect wealth accumulation and financial preparedness for retirement. However, there is limited prior evidence due to both data constraints and identification challenges. Measuring savings across all asset classes is difficult, as administrative microdata usually lack comprehensive information on household savings and wealth. Furthermore, credible identification of saving responses requires exogenous variation in mandate coverage.

This paper estimates the effects of mandatory pension plans on comprehensive measures of savings, leveraging administrative tax data from the Swiss canton of Bern and identifying variation from the occupational pension plan mandate. I begin by estimating the overall effect of mandatory retirement contributions on total savings. Then, I characterize the saving response throughout the portfolio. Detailed information on savings and wealth allows me to decompose the response into voluntary pension savings and private non-pension savings by asset type, including financial assets, business assets, property, other wealth, and debt. Finally, I examine how liquidity constraints influence workers' saving responses.

Switzerland is a particularly suitable laboratory for studying saving responses to mandatory pension plans as the rich administrative data and institutional context enable me to overcome the obstacles highlighted above. I draw on income tax and wealth tax records to construct granular measures of savings at the asset-type level. The universal coverage of the administrative data ensures representativeness and provides ample statistical power to precisely estimate effects. Measurement error and misreporting, common concerns with survey data, are limited because the tax authority verifies the tax records for administrative purposes. With savings data in hand, I approximate the ideal experiment exploiting the fact that the pension plan mandate only applies to workers with earnings above a threshold, providing compelling identifying variation in mandatory retirement contributions among otherwise very similar workers. The institutional setting in Switzerland is similar to many other high- and middle-income countries. The pension system consists of three pillars in the form of a pay-as-you-go scheme, employer-based pension plans, and preferentially taxed private pension accounts. This raises the likelihood that the findings of this paper extrapolate to other contexts.

The Swiss pension plan mandate requires employees whose annual earnings exceed a relatively low threshold of around CHF 20,000 (corresponding roughly to the 30^{th} percentile of the earnings distribution) to contribute to an employer-sponsored fully funded pension account.[1] Individuals subject to the mandate cannot opt out. Contributions are calculated by applying a legally defined age-specific contribution rate to qualifying earnings which equal gross earnings minus a deduction. For workers whose earnings are marginally above the mandate threshold, the contribution rate is applied to a fixed minimum. This causes a discontinuity in mandatory pension savings at the cutoff, meaning that otherwise similar workers are required to contribute substantially different amounts to employer-based pension accounts. In contrast to many public pension systems, these mandatory pension plans feature a strong and salient link between contributions and future benefits. Because being enrolled in an occupational pension plan raises the cap on contributions to preferentially taxed private pension accounts, there is also a discontinuity in the cap on private pension savings at the threshold.

I combine the sharp policy variation with high-quality administrative tax data on income, wealth, and savings from the second-largest Swiss canton, Bern, spanning years 2005 to 2017. The data cover the entire adult population and provide detailed information on workers' voluntary retirement savings, wealth, and debt, enabling me to construct savings measures by asset type. They also allow me to link partners to include household-level variables in the analysis. These data are quite exceptional as Switzerland is one of only three OECD countries that still levy a comprehensive wealth tax (OECD, 2018*c*).

To estimate the saving response to mandatory pension plans, I leverage the variation in mandate coverage provided by the earnings threshold within a regression

[1] For most of the sample period 2005–2017, the Swiss franc (CHF) was trading roughly at parity with the US dollar. Hence, I do not report separate figures in US dollar.

discontinuity framework. As workers with earnings just below and just above the threshold are very similar in all dimensions, except that only those above are subject to the mandate, this allows me to identify the causal effect. I estimate that the mandate causes a discontinuity in mandatory retirement savings of CHF 406, corresponding to around 2% of workers' earnings.

Despite the sizeable mandatory pension contributions, I find that the mandate's effect on total savings is limited. The point estimate is CHF 110 and implies a crowd-out rate of 73%. Based on the 95% confidence interval, I can reject increases in total savings by more than CHF 330. This implies that workers undo a large share of the impact of mandatory retirement contributions on overall savings by reducing other savings.

To identify the types of savings that drive the crowding-out response, I decompose the overall effect across the portfolio. I split up total savings into private savings, which include any type of savings not explicitly earmarked for retirement, and voluntary pension savings. I estimate separate effects for these two savings categories and then disaggregate them further to determine the saving response for each asset type. In sum, I find that workers offset mandatory retirement contributions by reducing private savings, especially in financial wealth. This implies that the mandate reallocates savings from more liquid financial assets to less liquid pension accounts.

More specifically, I estimate that the mandate lowers private savings by CHF 375. This drop is of similar magnitude as the mandated pension contributions. Breaking down private savings into asset types, I find that this decline is mainly driven by savings in financial assets which include bank accounts, stocks, bonds, and other investments. While I estimate a statistically significant decrease in financial savings by CHF 260, I do not find significant effects on other asset types such as business wealth, property wealth, other types of wealth, or debt, and the precision of most estimates allows me to rule out even modest changes. It seems intuitive that workers respond to the mandate mainly by lowering savings in financial assets as they are more liquid than other types of wealth.

While the mandate decreases private savings, I find a highly significant increase in voluntary pension savings by CHF 112. These consist of private pension savings and voluntary lump-sum buy-ins into occupational pension plans which I can both observe in the tax data because they are deductible from taxable income. The positive effect is surprising but it is confounded by the increase in the contribution cap on preferentially taxed private pension savings at the threshold. I separate out the impact of the changing cap from the effect of mandatory retirement contributions by estimating the effect across the distribution of private pension savings. I find evidence that the increase in private pension savings is entirely driven by the change in the contribution cap. Holding the cap fixed across the threshold, I estimate a precise null effect on private pension savings. In line with this result, I also find a precisely estimated null effect on occupational pension buy-ins.

As workers offset mandatory pension contributions by reducing savings in liquid assets, I investigate how liquidity constraints affect the saving response. I estimate heterogeneous effects across workers with different levels of baseline financial assets or household income, finding consistent results across these two proxies for liquidity. In line with standard models of saving under borrowing constraints, liquidity-constrained workers are less able to decrease their private savings to offset mandatory pension contributions. Consequently, the mandate increases their total savings by around CHF 200–350, representing 1–1.75% of earnings. Workers with high levels of financial wealth or household income respond to the mandate (and the associated increase in the cap on private pension contributions) by lowering private savings disproportionately to offset the mandated pension contributions and channel more of their savings into tax-advantaged private pension accounts, while keeping overall savings constant. Whereas the regression discontinuity approach provides a local effect estimate for workers near the mandate threshold, the full crowding-out among those with high household income suggests that crowd-out rates further up the earnings distribution are likely even larger than the 73% headline estimate.

This paper contributes to three strands of the literature on saving responses to pension policies. First, it adds to the thin evidence base on mandatory pension plans. While vast literatures study the savings effects of other types of pension policies such as financial incentives and automatic enrollment (see, e.g., Choi, 2015; OECD, 2018*a*), we lack systematic evidence on the saving responses to mandatory pension plan contributions. The few existing papers find weak to moderate crowding-out of other savings (Card and Ransom, 2011; Arnberg and Barslund, 2014; Chetty et al., 2014; Friedberg, Leive and Cai, 2024). Seminal work by Chetty et al. (2014) uses a regression discontinuity approach similar to this paper and finds little crowding-out. But the identifying variation in mandatory retirement savings is only around \$50 – an order of magnitude smaller than in the Swiss setting – and the threshold is at roughly \$5,000, so the marginal workers affected by the mandate have very low earnings and likely face liquidity constraints. Card and Ransom (2011) and Friedberg, Leive and Cai (2024) study the effect of mandatory pension contributions on voluntary retirement savings among faculty members at US universities. Card and Ransom (2011) find crowd-out rates of 60–80% for mandatory employee contributions but only half of that for employer contributions. Friedberg, Leive and Cai (2024) estimate a lower crowd-out rate of 30%. Because they rely on data from pension plan providers, neither of the two papers accounts for saving responses outside pension accounts, resulting in a lower-bound estimate of the overall crowd-out rate. In this paper, I estimate precise effects of mandatory pension contributions across all asset types, providing novel evidence of portfolio reallocation responses to pension policies. In particular, I document strong substitution between mandatory contributions and *non-pension* savings, which is not detected in prior work using pension plan data. By showing that liquidity constraints weaken the crowding-out response, I provide a potential explanation for the variation in savings effects across workers with different income levels in the literature.

Second, my findings are related to the literature on automatic saving policies, such as automatic enrollment and defaults, which aim to nudge workers into raising retirement savings. The key difference between mandatory pension plans and automatic saving policies is that mandates do not allow individuals to opt out, which constrains the set of available behavioral responses. The evidence consistently shows that automatic saving policies substantially increase pension plan enrollment and contributions in the short run (Madrian and Shea, 2001; Choi et al., 2004; Thaler and Benartzi, 2004; Beshears et al., 2009; Benartzi and Thaler, 2013; Chetty et al., 2014; Blumenstock, Callen and Ghani, 2018; Cribb and Emmerson, 2020; Chalmers et al., 2022; Bucher-Koenen, Wallossek and Winter, 2024). But recent research documents that workers undo part of these savings gains in the medium to long term, e.g., by contributing at lower rates at subsequent employers without automatic enrollment and withdrawing funds upon job separation (Derby, Mackie and Mortenson, 2023; Falk and Karamcheva, 2023; Choi et al., 2024; Choukhmane, 2024). These findings highlight potential advantages of mandates over nudges, as mandates prohibit workers from opting out or decreasing contribution rates below the statutory minimum (as well as limiting withdrawals). My results complement this evidence, showing that even strict mandate policies may fail to boost overall savings as workers respond by reducing private savings outside retirement accounts. But mandatory pension plans can still improve financial preparedness for retirement by reallocating savings from liquid private accounts to illiquid pension accounts which are more likely to remain untouched until retirement.

Whereas the early literature on auto-enrollment focused on retirement savings, recent work examines the effects on debt and financial assets outside pension accounts. Using detailed data from a UK bank, Choukhmane and Palmer (2024) find that a substantial share of increased retirement contributions is funded by reduced deposit balances and increased debt. This is consistent with Beshears et al. (2024) who show that employees of small UK businesses partly finance automatic savings with higher unsecured debt. By contrast, Beshears et al. (2022) do not find an effect on debt or credit scores of civilian employees of the US Army. I advance the emerging literature on the response of non-retirement savings to pension policies by estimating separate effects for each asset type, including but not restricted to financial wealth and debt.

Third, this paper relates to the longstanding literature on the savings effects of public pension systems, which also rely on mandatory contributions. The evidence suggests that private savings respond significantly to changes in public pension contributions and benefit eligibility, but the degree of substitution varies across settings and reforms (Feldstein, 1974; Attanasio and Brugiavini, 2003; Attanasio and Rohwedder, 2003; Bottazzi, Jappelli and Padula, 2006; Aguila, 2011; Lachowska and Myck, 2018; Lindeboom and Montizaan, 2020; Etgeton, Fischer and Ye, 2023; García-Miralles and Leganza, 2024). I show that mandatory pension plans, which are characterized by a stronger and more salient tax-benefit linkage than public pensions (Bozio, Breda and Grenet, 2023), generate strong crowding-out effects. While the literature on public pensions typically relies on survey data, I use high-quality administrative data to estimate saving responses throughout the portfolio, similar to García-Miralles and Leganza (2024) who study the effect of delayed public pension benefit eligibility on private savings.

The remainder of this paper is organized as follows. In Section 2, I discuss the relevant parts of the Swiss pension system. Section 3 describes the administrative tax data and data preparation. Section 4 explains the regression discontinuity approach that exploits the discontinuity in mandate coverage at the earnings threshold. Section 5 presents the main results on the saving responses to mandatory pension plans. Section 6 explores the role of liquidity constraints. Section 7 concludes.

2 Institutional Background

The Swiss old-age provision system consists of three pillars: (i) a standard pay-as-you-go system, (ii) a fully funded occupational pension system, which includes the mandatory pension plans studied in this paper, and (iii) voluntary private pension accounts. The structure is similar to pension systems in many other countries that consist of a government-backed defined-benefit plan, employer-sponsored defined-contribution plans, and preferentially taxed private pension accounts (e.g., in Denmark or the US, see Chetty et al., 2014).[2] This section introduces the relevant institutional context, focusing on the rules of the occupational pension system and voluntary private pension accounts from 2005 to 2017, the years used in the main analysis. There have been no major changes to the pension system during this period.[3] Appendix B describes the Swiss old-age provision system in more detail.

[2]Simplifying, the corresponding three pillars in the US are Social Security, 401(k) plans, and IRAs.

[3]The occupational pension system was reformed between 2004 and 2006, but the relevant changes were implemented by 2005. Appendix Section B.4 provides more information about the reform.

2.1 Occupational Pension System

The occupational pension system requires most workers to contribute to an employer-sponsored pension account. Employees must be enrolled in a pension plan by their employer if their annual gross earnings in the main job exceed a legally defined threshold. The pension savings mandate applies to women between 25 and 64 years and men between 25 and 65 years of age. The mandate is binding: employees meeting these conditions cannot opt out. Consequently, coverage is quite comprehensive. In 2017, 4.2 million individuals were enrolled in an occupational pension fund, representing around 83% of the Swiss labor force (Federal Statistical Office, 2019). Total contributions equalled CHF 54 billion, equivalent to 8% of GDP. These numbers highlight that occupational pension plans are one of the most important savings instruments in Switzerland.

The earnings threshold determining mandate coverage has increased gradually from CHF 19,350 in 2005 to CHF 21,150 in 2017.[4] While the vast majority of occupational pension plans use the statutory threshold, some may enroll employees with earnings below. I cannot directly investigate this because occupational pension contributions are not observed in the tax data. But Ecoplan (2010) show that there are only small discrepancies between the number of employees enrolled in occupational pension plans according to information from plan providers and the equivalent counts computed using earnings data collected by the social insurance system. Hence, I can reliably infer pension plan enrollment from workers' earnings.

Mandatory occupational pension savings are calculated by applying a contribution rate to qualifying earnings which equal gross earnings minus a deduction. The deduction has increased from CHF 22,575 in 2005 to CHF 24,675 in 2017 and serves a similar purpose as the mandate threshold. The statutory age-specific contribution rates are 7% for ages 25–34, 10% for ages 35–44, 15% for ages 45–54, and 18% for ages 55–64 among women and ages 55–65 among men. Employers must pay at least half the contribution. The share paid by employees is deducted from their earnings by the employer and directly transferred to the pension fund, along with the employer part. Contributions are exempt from income tax.

Importantly, there is a minimum for qualifying earnings. This has risen from CHF 3,225 to CHF 3,525 over the sample period. For employees whose earnings exceed the mandate threshold by only a small amount, the contribution rate is not applied to the marginal earnings above the threshold or deduction but to that min-

[4] Like most parameters of the occupational pension system, the threshold is set as a function of the maximum pension in the pay-as-you-go system in order to avoid overinsurance. Benefit levels in the pay-as-you-go system are usually adjusted every other year based on an index capturing the mean of nominal wage growth and inflation. Note that pay-as-you-go contributions are levied on total earnings, without a cap, even if they do not lead to higher benefits.

imum amount. This creates a discontinuity in mandatory pension contributions at the cutoff. I leverage this source of exogenous variation to study the effects of the mandate. Appendix Figure A.1 illustrates the calculation of mandatory retirement contributions as a function of gross earnings for a worker aged 45–54 in 2017.[5]

In addition to mandatory contributions, many occupational pension schemes allow lump-sum "buy-ins" which can also be deducted from taxable income.[6] These buy-ins can be directly observed in the tax records.

Workers are regularly informed about their contributions, investment returns, and the capital in their pension account. Upon retirement, the accumulated pension wealth can be withdrawn in the form of a lifelong annuity, a lump-sum payment, or a combination of the two. This underscores the strong and salient link between mandated pension contributions and future benefits. While contributions and returns on investment are exempt from income tax (and there is no capital gains tax in Switzerland) and occupational pension wealth is exempt from wealth tax, pension benefits are taxed.[7]

2.2 Private Pension Savings

In addition to contributing to the pay-as-you-go system and occupational pension plans, employees can make voluntary contributions to designated private pension accounts. These contributions can also be deducted from taxable income up to an annual cap. The access to those funds is restricted until individuals enter retirement. Private pension accounts need to be set up separately with a bank or insurance company. In the remainder of this paper, I refer to this savings vehicle as "private pension savings". Total private pension savings in designated accounts equal roughly CHF 10 billion per year, representing 1.5% of Swiss GDP (Schüpbach and Müller, 2019).

The contribution cap depends on whether workers are enrolled in an occupational pension plan. If they are enrolled, they can make yearly private pension contributions up to a fixed cap which has grown from CHF 6,192 in 2005 to CHF 6,768 in 2017. Workers not enrolled in an occupational pension plan can contribute up to 20% of their net earnings.[8] Importantly, this means that the cap is significantly lower for workers just below the mandate threshold who are not enrolled in an occupational

[5]Equation (4) in Appendix Section C.4 shows how to compute mandatory pension savings.

[6]Individuals can make buy-ins up to the level of savings that they would have accumulated, had they been earning their current salary since they were 25 years old (Kuhn, 2020).

[7]The majority of OECD countries applies a similar "exempt-exempt-taxed" regime (OECD, 2018*b*).

[8]There is also a fixed upper bound on private pension contributions for individuals not enrolled in an occupational plan which has increased from CHF 30,960 to CHF 33,840 over the sample period. This applies in particular to the self-employed, but it is not relevant for employees included in my analysis because the cap equal to 20% of net earnings binds first.

pension plan than for workers just above. Below but close to the threshold, the cap is between CHF 3,000 and CHF 4,000 in all years in the estimation sample.[9] Workers above the cutoff are required to be enrolled in an occupational pension plan, so they can all make private pension contributions up to the fixed cap between CHF 6,000 and CHF 7,000. This discontinuity in the contribution cap at the threshold could confound the estimated response of private pension savings to the mandate. In the empirical analysis, I carefully disentangle the impact of the change in the cap from the effect of mandatory retirement contributions.

There are strong tax incentives for private pension savings, in particular for high-income and high-wealth individuals, because contributions can be deducted from taxable income and the accumulated capital is exempt from the wealth tax. Upon retirement, private pension capital can be claimed as an annuity subject to income tax or as a lump sum taxed at special but usually preferential rates.

3 Data

Assessing the effect of the pension plan mandate on saving behavior requires comprehensive individual-level measures of savings. These are challenging to obtain because high-quality microdata on assets and savings are scarce. Switzerland is a particularly well-suited setting to study saving choices and wealth accumulation due to the availability of administrative tax data on wealth.[10] However, there is no individual-level dataset on wealth with nationwide coverage because the wealth tax is only collected at the cantonal and municipal level.

I draw on administrative tax microdata from the canton of Bern which is the second-largest Swiss canton by population and approximately representative for Switzerland (Brunner, Meier and Näf, 2020).[11] The tax records provide detailed and comprehensive information on income, wealth, and savings. They also contain basic demographic information including the year of birth, gender, marital status, the number of children, and the municipality of residence. The dataset is an annual panel covering all taxpayers in the canton between 2002 and 2017. It is based on the tax returns that all permanent residents must file. Accordingly, the composition of individuals in the data only changes due to migration or death. Full coverage

[9] The mandate threshold is around CHF 20,000 in gross earnings and the employee share of social insurance contributions a bit above 6%, with the exact numbers depending on the year. Therefore, the simplified calculation for the lower contribution cap is: CHF 20,000 $\times (1 - 0.06) \times 0.2 =$ CHF 3,760. The cap can be higher if workers have additional earnings, e.g., from working a second job.

[10] Since France replaced its wealth tax with a tax on real estate in 2018, only two OECD countries other than Switzerland still levy a wealth tax: Norway and Spain (OECD, 2018*c*).

[11] According to population statistics from the Federal Statistical Office, 1.03 million people were resident in the canton of Bern in 2017: https://www.bfs.admin.ch/bfs/en/home/statistics/population.html [accessed on 22 October 2021].

of the population is key both for representativeness and to have enough statistical power when zooming in on the group of workers around the mandate threshold in the regression discontinuity analysis. As taxes are levied at the household level in Switzerland, I can link individuals to their partners if they are married or in a civil union. This allows me to include household-level variables in the analysis. The tax records are verified by the tax authority for administrative purposes, limiting the extent of measurement error and misreporting.

Because the pension plan mandate applies at the individual level while the tax unit in Switzerland is the household, I split up married couples into individual observations. Most of the important variables for the analysis, including earnings from work, private pension savings, and occupational pension buy-ins, are reported at the individual level in the tax data. For all income and wealth components reported only at the household level, I equally assign half to each partner (following Fagereng et al., 2020). Appendix C describes the dataset, data cleaning and preparation, and variable construction in more detail.

3.1 Income

The tax records provide detailed information on all types of income. This includes earnings from employment separately for the main job and other jobs, which is crucial for constructing the earnings measure determining whether workers are subject to the pension plan mandate. The data also provide information on self-employment income, business income, financial income, property income, transfer income, and pension income.

The relevant earnings concept for the pension plan mandate is gross earnings in the main job, before deducting social insurance and occupational pension contributions. This is the running variable in the regression discontinuity approach. Because I observe main-job *net* earnings in the tax data, I build a calculator using the year-specific social insurance and occupational pension contribution schedules to back out gross earnings for each individual. While this is straightforward for most individuals, gross earnings narrowly below and above the mandate threshold can result in the same net earnings recorded in the tax data, because workers above the threshold have the employee part of mandatory pension contributions deducted whereas those below do not. Thus, for a small number of individuals within at most CHF 350 of the threshold, I cannot unambiguously impute gross earnings. I exclude these individuals from the main analysis.

3.2 Wealth

The tax data also provide very comprehensive information on wealth which I use to construct savings measures by asset type. This includes financial wealth, business wealth, property wealth, other types of wealth, and debt. Financial wealth captures bank accounts, stocks, bonds, and any other types of financial investments. Property wealth includes any property people own, including primary residences, but the valuation is typically below the true market value and only updated periodically.[12] "Other wealth" includes the value of an individual's share of assets held through joint heirship or other forms of co-ownership and any assets not included elsewhere such as cash, gold, valuable art, cars, boats, horses, etc. Debt covers all types of private debt such as mortgages, loans, overdrafts, and credit card debt.

The only type of wealth missing from the data is pension wealth held in occupational and private pension accounts as these are exempt from the wealth tax. This is of minor concern for my analysis as I can observe voluntary occupational and private pension savings in the tax data and impute mandatory occupational pension savings based on individuals' gross earnings and the relevant contribution schedule.

3.3 Savings

To study the effects of mandatory pension plans on the entire savings portfolio, I obtain measures of savings for all types of assets. Whereas some savings measures are directly observable in the tax records, others can be imputed based on available information. Mandatory occupational pension contributions are not included in the data because they are withheld at source by the employer, so I impute these by applying the statutory contribution schedule to gross earnings. I define mandatory pension savings as the sum of the employee and employer contribution as empirical evidence suggests the incidence is likely on workers due to the strong tax-benefit linkage (Bozio, Breda and Grenet, 2023). Voluntary retirement savings such as lump-sum buy-ins into occupational pension plans and contributions to private pension accounts are directly observed in the tax records because they can be deducted from taxable income.

For all other types of savings not explicitly earmarked for retirement, I construct annual savings measures using the wealth data. As is common in the literature (e.g., Chetty et al., 2014; García-Miralles and Leganza, 2024), I compute savings as the change in the reported value of assets in year t relative to year $t-1$. This measure includes both "active savings" and "passive savings" from price changes (see Fagereng et al., 2021). This is conceptually attractive as it is the most comprehensive

[12] As a rule of thumb, the tax value of property corresponds to around 60% of the market value (OECD, 2018*c*).

savings measure in line with the classic Haig-Simons income concept (Aguiar, Moll and Scheuer, 2024). But in practice, it means that private savings are volatile due to fluctuations in asset prices.[13]

To estimate the effects of the mandate on more aggregated measures of savings, I compute three main outcomes: (i) overall private savings are calculated as the change in net wealth relative to the previous year, which takes into account all types of non-pension assets and debt; (ii) overall voluntary retirement savings are defined as the sum of private pension contributions and occupational pension buy-ins; (iii) total savings are computed as the sum of overall private savings, overall voluntary retirement savings, and mandatory occupational pension contributions. To reduce the variance stemming from asset price volatility, I follow García-Miralles and Leganza (2024) and winsorize overall private savings and asset-type-specific private savings at percentiles 5 and 95 of the distribution in the main estimation sample.[14]

4 Empirical Strategy

Estimating the causal effect of mandatory pension plans on savings is challenging because one needs to disentangle the saving response to the mandate from unobserved heterogeneity in saving preferences. The ideal, but impractical, experiment would force randomly selected individuals to contribute to retirement accounts. I approximate this using a regression discontinuity design that leverages the earnings threshold of the Swiss pension plan mandate which requires otherwise similar workers to contribute substantially different amounts to employer-based pension accounts. The sharp mandate threshold provides credibly exogenous variation in mandate coverage that I can use to identify its causal effect. Combined with detailed and comprehensive tax records, this allows me to examine the saving response across the entire portfolio.

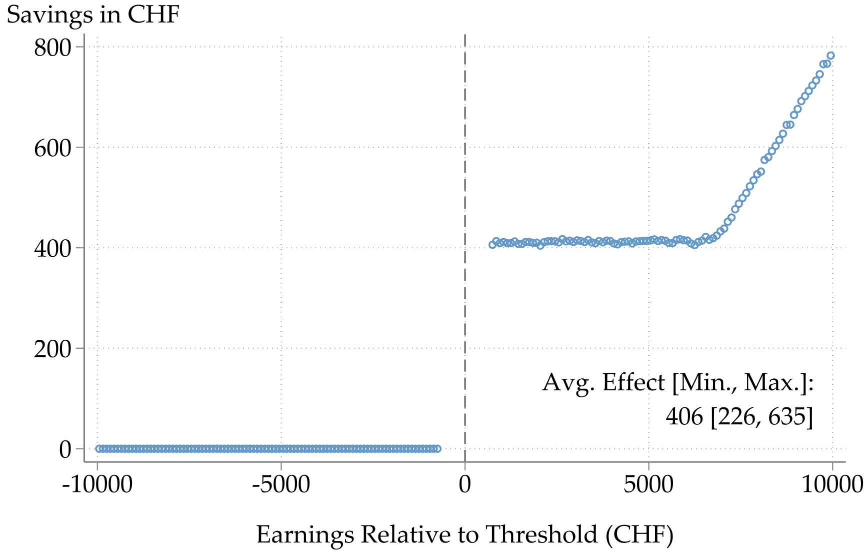
4.1 Discontinuity in Mandatory Pension Contributions

Employees with earnings above the mandate threshold automatically contribute a fraction of their salary to an occupational pension account, while those below are not required to make those savings. For those marginally above the threshold, the contribution rate is applied to the minimum qualifying earnings. Thus, annual mandatory pension savings jump from zero to CHF 225–635 at the cutoff, with the exact amount

[13] I cannot strip out asset price changes because data at the level of individual assets or transactions are not available.

[14] The exceptions are savings in business wealth and property wealth which I winsorize at percentiles 2.5 and 97.5 because they are rare for workers in the estimation sample. Only 9.5% and 7.9% of worker-year observations have non-zero business savings or property savings, respectively.

FIGURE 1: EFFECT OF PENSION PLAN MANDATE ON MANDATORY PENSION SAVINGS



Notes: Regression discontinuity plot showing the effect of the pension plan mandate on mandatory occupational pension savings using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. Occupational pension savings are predicted by applying the statutory contribution schedule to gross earnings in the main job following Equation (4). See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

depending on the age-specific contribution rate and the year-specific minimum qualifying earnings. Figure 1 depicts the mechanical effect of the mandate on average mandatory pension savings. There is a discontinuity of CHF 406 at the threshold, corresponding to roughly 2% of earnings. I leverage this variation in mandate coverage using a sharp regression discontinuity (RD) design to estimate the effect of the mandate on saving behavior.

4.2 Regression Discontinuity Design

Identification. Assuming that workers do not adjust their earnings to sort below or above the mandate threshold, the cutoff generates variation in treatment assignment that is orthogonal to other individual characteristics. The fundamental idea is that workers just below the threshold can be used as a valid counterfactual for those just above the threshold under the assumption that they are comparable in all relevant dimensions except for whether they are subject to the mandate.

Formally, identification requires that the (average) potential outcomes are continuous across the threshold (Hahn, Todd and Van der Klaauw, 2001). The continuity assumption can be expressed in the following way. Let $Y_{it}(1)$ and $Y_{it}(0)$ denote the savings outcome of interest for individual i in year t who is subject or not subject to the mandate, respectively. The sharp RD estimand τ_{RD} identifies the local average treatment effect on individuals with near-threshold earnings X_{it}, provided that the conditional expectation functions are continuous in x at the cutoff c_t:

$$\tau_{RD}(c_t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0)|X_{it} = c_t] = \lim_{x \downarrow c_t} \mathbb{E}[Y_{it}|X_{it} = x] - \lim_{x \uparrow c_t} \mathbb{E}[Y_{it}|X_{it} = x] \quad (1)$$

The continuity assumption could be violated if workers or their employers were willing and able to manipulate earnings strategically and precisely to impact workers' treatment status. In this case, potential discontinuities in observable and unobservable characteristics at the cutoff could confound estimates of the causal effects of the mandate. I discuss and test the validity of the identifying assumption in Section 4.3.

Estimation. Following the standard advice of the methodological RD literature (see in particular Gelman and Imbens, 2019), I estimate the target parameter τ_{RD} by $\hat{\beta}_1$ from a local linear regression, allowing the slope of the control functions to vary on each side of the cutoff. The main specification is given by

$$Y_{it} = \beta_0 + \beta_1 \times \mathbb{1}\{X_{it} \geq c_t\} + \beta_2(X_{it} - c_t) + \beta_3(X_{it} - c_t) \times \mathbb{1}\{X_{it} \geq c_t\} + Z'_{it}\gamma + \epsilon_{it}, \quad (2)$$

where Y_{it} is a savings outcome of interest for individual i in year t, X_{it} denotes earnings in the main job (the running variable), c_t is the mandate cutoff determining treatment assignment, and ϵ_{it} is the error term. Z_{it} represents a vector of controls consisting of year fixed effects, age fixed effects, gender, and marital status. Control variables are not necessary for identification, but their inclusion improves precision (Calonico et al., 2019). The treatment variable is $\mathbb{1}\{X_{it} \geq c_t\}$ which indicates being subject to the pension plan mandate. Provided that ϵ_{it} does not change discontinuously at the threshold, $\hat{\beta}_1$ provides an unbiased estimate of the average treatment effect of the mandate on savings outcome Y_{it} for workers in proximity of the cutoff.

For ease of interpretation and to keep the effective estimation sample consistent, I use a bandwidth of CHF 10,000 for all outcomes in my main analysis.[15] To give more weight to observations close to the cutoff, I use a triangular kernel. To account for potential sorting around the threshold, the main analysis employs a "donut hole" approach excluding workers who have earnings within CHF 700 of the threshold (Bajari et al., 2011; Barreca et al., 2011; Barr, Eggleston and Smith, 2022). This amount

[15] I have also computed mean squared error optimal bandwidths using the data-driven bandwidth selector proposed by Calonico, Cattaneo and Titiunik (2014*a*,*b*). However, their procedure suggests very small bandwidths that leave few observations for estimation and result in large standard errors. Instead, I use a consistent bandwidth across all outcomes and verify the robustness of the results with respect to the bandwidth choice in Section 5.5.

corresponds to the maximum mandated pension contribution for workers just above the threshold.[16] This means that even if workers perceive the entire economic incidence of both the employee and employer contribution to be on them (as suggested by Bozio, Breda and Grenet, 2023), they would still prefer to have gross earnings above the threshold and be subject to the mandate over gross earnings more than CHF 700 below the cutoff. Hence, for workers remaining in the sample, manipulation of earnings to affect treatment status is very unlikely. This restriction also takes care of the issue that gross earnings of workers within CHF 350 of the threshold cannot be imputed unambiguously (see Appendix Section C.2).[17]

Inference. I compute standard errors using the heteroskedasticity-robust nearest-neighbor variance estimators proposed by Calonico, Cattaneo and Titiunik (2014*a*) and implemented in statistical software by Calonico, Cattaneo and Titiunik (2014*b*) and Calonico et al. (2017), as they are typically more robust in finite samples than conventional Huber-Eicker-White heteroskedasticity-robust standard errors (Calonico, Cattaneo and Titiunik, 2014*a*).

Sample Restrictions. The main analysis uses data from 2005 to 2017. I exclude 2002–2004 because the 2005 reform changed a number of provisions related to the mandate, including substantially lowering the threshold and changing the contribution rates for women (see Appendix Section B.4 for more information on the reform). As the pension plan mandate only applies to workers from the age of 25, I restrict the estimation sample to those aged 25–60.[18] Further, I exclude anyone receiving more than CHF 25,000 in self-employment income because individuals whose main source of income is self-employment can request to be exempt from the mandate even if they have some earnings from employment. Finally, I remove individuals who are not in the data in the preceding year because private savings cannot be computed for them (4% of the sample, most likely people moving to the canton from elsewhere). This leaves me with an effective estimation sample of 597,120 observations from 191,400 unique individuals.

Appendix Table A.1 presents summary statistics for the analysis sample. The average worker is 43 years old. 82% of workers in the sample are women, 67% are married. The average worker receives CHF 25,300 in total income, most of which

[16] During the sample period, the maximum mandatory pension contribution for workers just above the threshold is: 18% $\times$ CHF 3,525 = CHF 634.50

[17] Estimation is performed using the Stata package `rdrobust` developed by Calonico, Cattaneo and Titiunik (2014*b*); Calonico et al. (2017).

[18] I impose a lower maximum age threshold than the legal retirement age – 62 years (2002–2004) and 64 years (2005–2017) for women, 65 years for men – to keep the age composition consistent across time and genders and to exclude people who retire early. Early retirement is widespread in Switzerland (Dorn and Sousa-Poza, 2005). I use the age group 20–24 years in a placebo check described in Section 4.3.

represents earnings from their main job. Mean total savings are CHF 4,000 per year. Private savings account for 63%, whereas voluntary pension savings account for 30% of total savings. Mandatory pension savings make up 6% of total savings, on average, but more than 10% among workers above the mandate threshold.

Robustness Checks. In Section 5.5, I demonstrate that the estimation results are robust to making different operationalization choices regarding the bandwidth, size of the donut hole, kernel function, control variables, winsorization, and years of data included in the estimation sample.

4.3 Validity of the Identifying Assumptions

The key identifying assumption is that there is a discontinuity in mandatory pension contributions at the threshold, whereas average potential outcomes are continuous. The main threat to identification stems from employees or employers manipulating earnings to sort around the threshold. The Swiss government usually announces the mandate threshold between the middle of September and the middle of October before it comes into force on the 1st of January. To affect their treatment status, employees would need to adjust their working hours, negotiate a change in the wage rate, or switch into a new job. Taking on a second job would not change treatment status because the mandate threshold applies to earnings in the main job only. Employers might try to push the earnings of their employees below the threshold in order to avoid paying the employer share of occupational pension contributions (assuming some of the incidence falls on them).

To rule out sorting around the threshold, the main analysis excludes workers who have earnings within CHF 700 of the threshold. As this exceeds the maximum mandatory retirement contribution of workers just above the threshold, endogenous sorting is unlikely to occur outside that range. Nevertheless, I assess the plausibility of the identifying assumptions by examining the continuity of the density of the running variable and predetermined covariates across the threshold, and by estimating saving responses for a placebo sample of workers aged 20–24 who are too young to be subject to the mandate.

Density Test. Following McCrary (2008), I test for a discontinuity in the density of earnings at the threshold. Panel A of Appendix Figure A.2 shows the frequency distribution of annual earnings in the main job within the bandwidth. Workers with earnings within CHF 700 of the threshold, who are excluded from the main analysis, are represented by the light blue bars. Note that I remove workers with gross earnings within CHF 350 of the threshold because their gross earnings cannot be imputed unambiguously (see Section C.2).[19]

The density looks smooth and has a similar level below and above the mandate threshold. There is not much indication of bunching around the cutoff. The number of workers just below the threshold is slightly higher than just above, but the difference appears small in magnitude relative to the overall frequencies. I test formally for the presence of a discontinuity employing the density test proposed by Cattaneo, Jansson and Ma (2018, 2020), using a local linear regression as throughout the RD analysis. As Panel B shows, the null hypothesis of a continuous density cannot be rejected, with a p-value of 0.65. This is in line with previous research finding that employees fail to precisely sort around earnings thresholds to take advantage of financial incentives (Chetty et al., 2011; Chetty, Friedman and Saez, 2013; Chetty et al., 2014).

Continuity of Predetermined Characteristics. As endogenous sorting around the cutoff would likely result in imbalances on covariates, I assess the continuity of a set of predetermined characteristics across the threshold. I implement this balance test by estimating the main RD specification in Equation (2) without control variables, using age, gender, marital status, number of children, or household income as the dependent variable.

Appendix Figure A.3 displays the results. I find that workers with earnings marginally below and above the threshold have very similar characteristics. Based on the precision of the estimates, I can rule out even small discontinuities for gender, marital status, number of children, and household income. I do find a significant discontinuity in age, implying that workers just above the threshold are on average a quarter of a year younger than those narrowly below. While the magnitude of this difference is modest, I address this potential source of bias by controlling for age fixed effects in the main analysis.

Placebo Sample. Based on the density test and smoothness of covariates across the threshold, endogenous sorting does not appear to be a major concern. But continuity of the potential outcomes can also be violated if there are other discontinuities at the threshold that affect savings, e.g., if the same cutoff value is used in another policy (Imbens and Lemieux, 2008). Ex ante, this seems highly unlikely as I am not aware of any other policies targeting the same cutoff and it moves from year to year between specific non-round numbers, defined as the part of the salary that is covered by the pay-as-you-go old-age insurance system (see Appendix B for more detail).

I can test this empirically, taking advantage of the fact that the mandate only affects workers from the year in which they turn 25. In a placebo exercise, I restrict

[19] This affects 16,607 individual-year observations, representing 2.6% of workers within the estimation window.

the sample to workers aged 20–24 and re-run the main regression discontinuity estimation. As these workers are not subject to the mandate, I should not find a saving response in the data. Appendix Figure A.4 shows the estimated placebo effects on total savings, overall private savings, and overall voluntary pension savings. In constrast to the main worker sample, I estimate precise null effects on all three savings outcomes among this group of young workers unaffected by the mandate. This is evidence that there is no other discontinuity at the threshold other than the mandate, at least none that affects both young workers and older workers.

5 Saving Responses to Mandatory Pension Plans

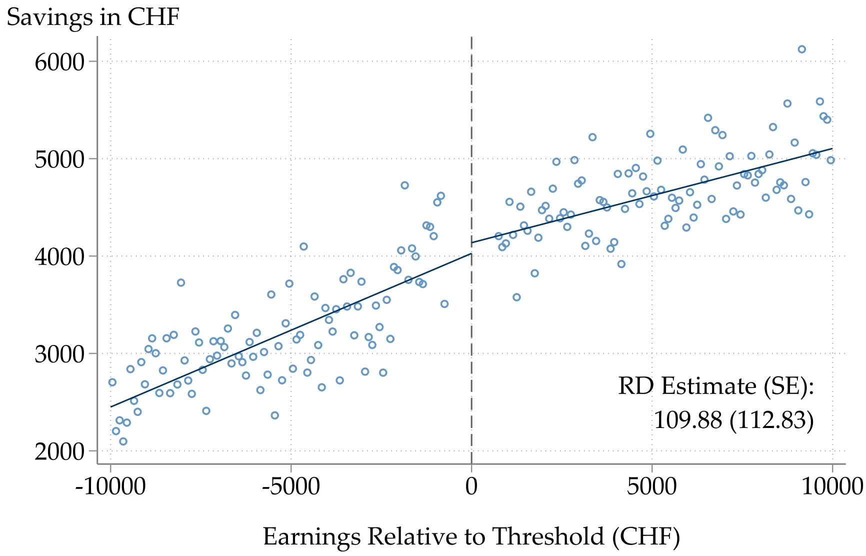
As described above, the pension plan mandate forces the average worker narrowly above the threshold to contribute around CHF 400 per year – close to 2% of their earnings – to an employer-sponsored pension account. This section presents the results from the regression discontinuity analysis leveraging this variation in mandatory retirement contributions. I begin by presenting estimates of the mandate's effect on total savings. Then, I investigate potential portfolio reallocation responses, estimating separate effects on private savings, which include any form of savings not specifically earmarked for retirement, and voluntary pension savings. To identify the specific asset types driving the response, I split these savings categories further into its components. Putting all the estimates together, I characterize the saving response to mandatory pension plans across the entire portfolio. Finally, I conduct a range of robustness checks to probe the sensitivity of my results.

5.1 Total Savings

Figure 2 plots total savings against earnings in the main job (i.e., the running variable), providing graphical evidence of the mandate's effect on overall savings. The dots are local sample means of 100 non-overlapping, evenly spaced bins on each side of the cutoff. The lines are linear fits from a non-parametric regression using all underlying datapoints and a triangular kernel. The point estimate and standard error for the treatment effect are displayed in the figure, obtained from estimating the RD specification in Equation (2).

Strikingly, there is no clear discontinuity at the cutoff, suggesting that the mandate has little impact on the overall savings level of workers. The point estimate of CHF 110 is not statistically significantly different from zero and corresponds to a crowd-out rate of 73%. Based on the 95% confidence interval, I can rule out increases in total savings by more than CHF 330. The headline estimate implies that mandatory retirement contributions are substantially offset by reductions in other types of sav-

Figure 2: Effect of Pension Plan Mandate on Total Savings



Notes: Regression discontinuity plot showing the effect of the pension plan mandate on total savings using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cut-off; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. Total savings are the sum of mandatory pension savings, private savings, and voluntary pension savings. Private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

ings, providing evidence of strong portfolio reallocation responses to the mandate. I proceed by decomposing the effect to reveal the type of savings that workers adjust to undo the mandate's impact.

5.2 Private Savings

Figure 3 displays the effects of the pension plan mandate on overall private savings and its components.

Overall Private Savings. Panel A plots the response of overall private savings. I find that private savings drop by CHF 375 at the cutoff, which is statistically highly significant. The magnitude of this effect is close to the average mandatory retirement contributions of roughly CHF 400 among workers above the threshold. This implies that workers respond to the mandate by lowering private savings which largely offsets its effect on overall savings.

Private Savings by Asset Type. Drawing on detailed information on individual wealth included in the tax data, I can investigate what types of assets account for

Figure 3: Effects of Pension Plan Mandate on Private Savings

Notes: Regression discontinuity plots showing the effect of the mandate using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. All outcomes are winsorized at percentiles 5 and 95, except savings in business wealth and property wealth which are winsorized at percentiles 2.5 and 97.5.

Source: Author's calculations based on administrative tax data from the canton of Bern.

the large drop in private savings. Panels B through F present the effect of the mandate on savings in financial assets, business assets, property, other types of wealth, and debt, respectively.[20] The results indicate that the overall drop is mainly driven by reduced savings in financial assets. As Panel B shows, the mandate lowers financial savings by CHF 260, again statistically highly significant. Financial assets include bank accounts, stocks, bonds, and other investments. While I cannot decompose financial assets further to probe whether the drop in savings is driven by checking accounts or more long-term investments, it seems plausible that workers use more liquid forms of wealth to fund the mandatory retirement contributions.

Panels C through F show that the mandate does not affect savings in business wealth, property wealth, other types of wealth, or debt. Three out of four point estimates are negative but they are not statistically significantly different from zero. For savings in property, I find a decrease by CHF 69 but the confidence interval is relatively wide because property investments are lumpy. The effect estimates for business wealth, other wealth, and debt are an order of magnitude smaller than the effect on financial savings and the precision of these estimates allows me to reject even small changes. This indicates that their response to the mandate is quantitatively negligible. The lack of an effect on debt is in line with Beshears et al. (2022) who do not find an impact of automatic enrollment on debt or credit scores in the US, but at odds with Beshears et al. (2024) and Choukhmane and Palmer (2024) which both find that workers in the UK partially finance higher pension savings induced by automatic enrollment through an increase in debt.

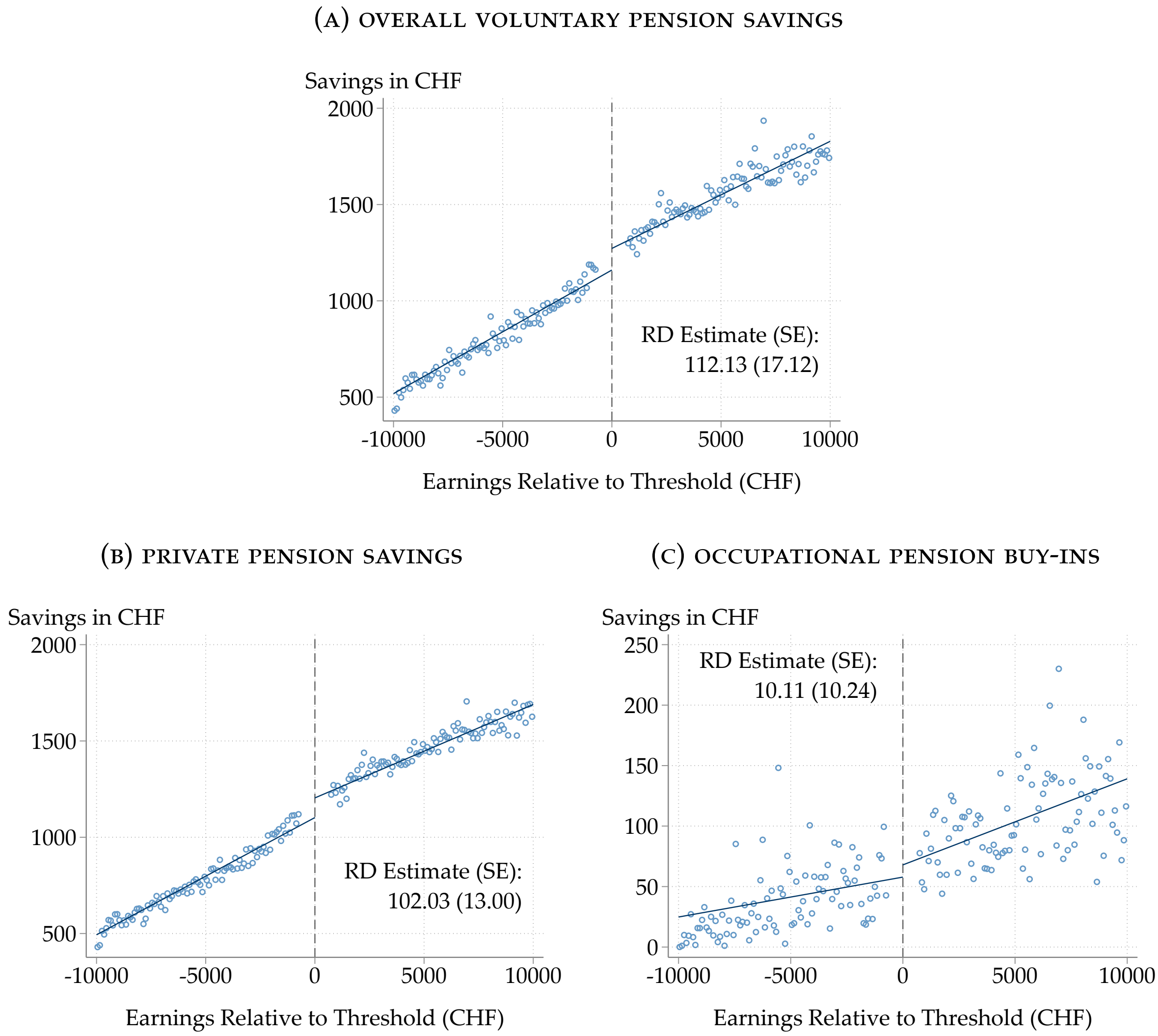
5.3 Voluntary Retirement Savings

Besides reducing private savings, the pension plan mandate could also affect workers' voluntary retirement contributions. Because they can be deducted from taxable income, I can observe both voluntary lump-sum buy-ins into occupational pension plans and private pension savings in separate accounts in the tax data. Below, I first estimate the effect on the sum of the two types of voluntary retirement savings before analyzing the response of each of them separately.

Overall Voluntary Pension Savings. Panel A of Figure 4 shows that workers respond to the mandate by *increasing* their overall voluntary retirement savings by CHF 112. The lower bound of the 95% confidence interval is CHF 79. While voluntary pension savings generally increase linearly with earnings, with similar slopes below and above the cutoff, there is a clear jump upwards at the threshold. The positive sign of the effect is surprising, as mandatory and voluntary retirement savings seem more

[20]"Other Wealth" includes an individual's share of assets in joint heirship or other types of co-ownership as well as other assets such as cash, gold, art, boats, etc.

Figure 4: Effects of Pension Plan Mandate on Voluntary Pension Savings



Notes: Regression discontinuity plots showing the effect of the pension plan mandate on voluntary pension savings using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

likely to be substitutes than complements. But the effect is confounded by the change in the contribution cap on private pension savings at the threshold. Below, I disentangle the impact of mandatory retirement contributions from the effect of the cap.

Private Pension Savings. Panel B documents that the mandate increases private pension savings by CHF 102. Clearly, this is driving the entire effect on overall voluntary pension savings. Because both mandatory pension savings and the cap on these preferentially taxed private pension savings increase discontinuously at the mandate threshold, this estimate reflects a combination of their potential effects. As explained in more detail in Section 2.2, workers just below the threshold face an annual cap on private pension savings of CHF 3,000–4,000, whereas those just above the threshold can contribute up to CHF 6,000–7,000, with the exact amount depending on the year.

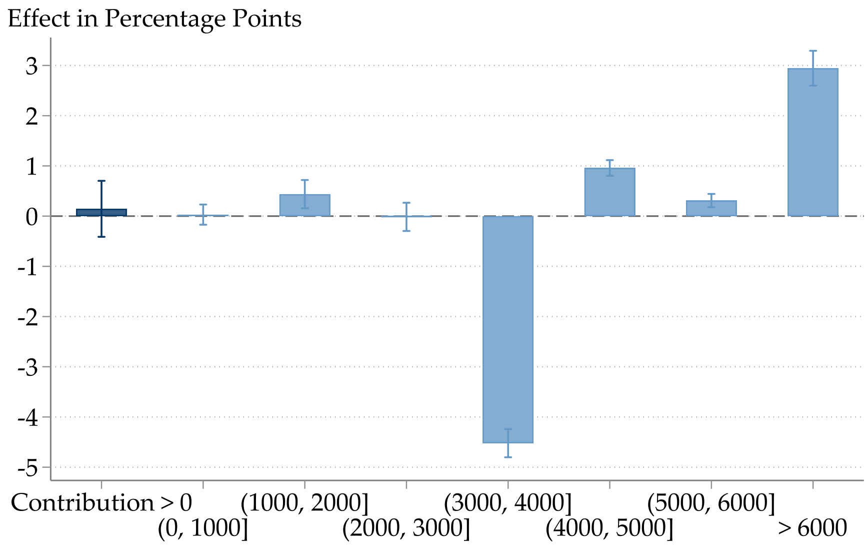
I disentangle the effect of mandatory pension contributions from the contribution cap in two separate exercises: First, I estimate the effects of the mandate along the distribution of private pension savings, exploiting that the two forces should have distinct impacts on different parts of the distribution: while mandatory pension savings should, if anything, decrease voluntary pension savings across the entire distribution, including at the extensive margin, the higher contribution cap should raise voluntary pension savings, but only at the upper end of the distribution where the cap is binding. In the second exercise, I isolate the effect of mandatory pension contributions by imposing the same contribution cap for workers above and below the threshold and re-estimating the effect on private pension savings.

Figure 5 presents the results of the first exercise, showing estimated effects on the share making non-zero private pension contributions (extensive margin) in dark blue and the share contributing amounts in intervals of CHF 1,000 in light blue. Appendix Figure A.5 displays the corresponding RD plots.[21] The results suggest that the effect of the mandate on private pension savings is entirely driven by the increase in the contribution cap at the mandate threshold. At the extensive margin and at the lower end of the distribution, where the contribution cap is irrelevant, effect estimates are close to zero. This suggests that there is not much substitution between mandatory and voluntary pension savings. By contrast, the share of workers with private pension savings in the interval where the cap below the threshold lies (CHF 3,000–4,000) drops by 4.5 percentage points. Conversely, the share contributing more than CHF 4,000 increases by almost 4.5 percentage points. This can be interpreted as the share of workers near the threshold for whom the lower contribution cap is binding. The share contributing CHF 4,000–5,000 increases by 1 percentage point, the share contributing CHF 5,000–6,000 by 0.3 percentage points, and the share contributing more than CHF 6,000 by 3 percentage points at the cutoff.

The results of the second exercise confirm that there is little substitution between mandatory pension contributions and private pension savings. As Appendix Figure A.6 shows, I estimate a precise null effect when limiting private pension contributions of all workers to at most 20% of their net earnings and re-estimating the effect of the mandate on this capped outcome.

[21] The RD plots in Appendix Figure A.5 clearly document the bunching of workers below the threshold at the lower contribution cap. For example, Panel E shows that a sizeable share of workers with earnings just below the cutoff (i.e., with gross earnings of around CHF 20,000) make contributions of CHF 3,000–4,000 which is where their cap is. This share is much lower above the cutoff where the cap is higher.

Figure 5: Effect of Mandate on Distribution of Private Pension Contributions



Notes: Coefficient plot showing the effect of the pension plan mandate for different intervals of private pension savings using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

Occupational Pension Buy-ins. Panel C of Figure 4 depicts the effect of the pension plan mandate on voluntary buy-ins into these plans. I also find a precisely estimated null effect for this type of voluntary pension savings, allowing me to rule out changes by more than CHF 30 at the 5% level. For workers in the income range considered here, buy-ins are of minor importance in general as only 0.7% of the worker-year observations in the estimation sample make non-zero buy-ins. Conditional on making a buy-in, the mean amount is CHF 9,700.

5.4 Summarizing the Saving Response Across the Portfolio

Mandatory pension plans have a modest effect on total savings, on average, as workers offset mandatory contributions by lowering their private savings. This allows them to maintain their consumption level despite a sizeable increase in forced savings. At the same time, there is no impact on voluntary pension savings, once the contribution cap is held fixed. This leads to a reallocation of savings from more liquid financial assets, presumably bank accounts as low-income workers are unlikely to hold significant amounts of stocks and bonds, to less liquid retirement accounts. Savings in other non-pension assets are not significantly affected by the mandate.

5.5 Robustness Checks

In this section, I check the sensitivity of the estimated effects on the main outcomes – total savings, overall private savings, and overall voluntary retirement savings – to different choices regarding the RD specification in Equation (2) and its operationalization. I show robustness with respect to changing the bandwidth, size of the donut hole, kernel weights, set of control variables, winsorization cutoffs, and the years of data included in the estimation sample.

Bandwidth. Appendix Figure A.7 plots point estimates and 95% confidence intervals for the effects of the mandate, varying the bandwidth symmetrically around the main bandwidth of CHF 10,000 – from CHF 2,000 to CHF 18,000. Overall, the findings are robust to using alternative bandwidths. Panel A documents that all bandwidths yield estimated effects on total savings that are close to zero and statistically insignificant, except for the CHF 4,000 bandwidth, which produces a negative effect estimate. Thus, there is no bandwidth at which the mandate is found to significantly increase total savings. As Panel B shows, I estimate a significant negative effect on private savings for all bandwidths except the narrowest, which yields a point estimate comparable to the main result but with a substantially wider confidence interval that includes zero. Panel C shows that the positive effect on voluntary pension savings remains relatively stable for bandwidths above CHF 6,000. At smaller bandwidths, the point estimates decline and the standard errors increase, making the estimates statistically indistinguishable from zero.

Size of Donut Hole. To rule out sorting around the threshold, the main analysis excludes workers within CHF 700 of the cutoff, as sorting is implausible to occur beyond that range (see Section 4.2). Appendix Figure A.8 examines the sensitivity of the estimates to the width of the donut hole that is removed from the estimation sample (Bajari et al., 2011; Barreca et al., 2011; Barr, Eggleston and Smith, 2022). When varying the donut hole from zero to CHF 1,400, the estimates remain consistent with the main results. The estimated effects on total savings (Panel A) and private savings (Panel B) are very stable for donut sizes up to CHF 1,000. They increase modestly for larger donut holes, but all confidence intervals continue to include the main point estimate. Panel C shows that the estimated effects on voluntary pension savings are always positive and highly statistically significant.

Kernel. Appendix Figure A.9 shows that the results are robust to using alternative weighting functions in the local linear regression. For all outcomes, using a uniform or Epanechnikov kernel yields point estimates and confidence intervals that are very similar to those obtained with a triangular kernel.

Control Variables. Appendix Figure A.10 displays the saving responses estimated using different sets of controls. Including only year fixed effects, or omitting controls entirely, yields point estimates and confidence intervals that are very similar to those from the main specification, which controls for year fixed effects, age fixed effects, marital status, and gender. This also shows that the controls do not improve the precision of the estimates by much.

Winsorization. Following García-Miralles and Leganza (2024), I winsorize private savings at percentiles 5 and 95 in the main analysis. To verify that the findings are not sensitive to the winsorization cutoffs, Appendix Figure A.11 compares the main results to estimates obtained when winsorizing the outcomes at percentiles 1 and 99 or percentiles 10 and 90 (as in Chetty et al., 2014), respectively. Although the width of the confidence intervals naturally varies with the degree of winsorization, the different estimates consistently point to sizeable reductions in private savings which limit the effect of mandatory pension plans on total savings.

Sample Period. The main analysis sample excludes years 2003 and 2004 because a reform implemented in 2005 lowered the mandate threshold and changed the contribution schedule for women.[22] As shown in Appendix Figure A.12, including these years in the estimation sample yields very similar results. The estimates for total savings and private savings are virtually unchanged. The positive effect on voluntary pension savings is slightly attenuated but remains sizeable and highly significant. This is expected, as the discontinuity in the contribution cap at the threshold, which drives the entire effect, was smaller prior to the reform.[23]

6 Saving Responses Under Liquidity Constraints

The average worker responds to mandatory pension contributions by sharply cutting private savings, especially in financial assets. But workers with limited liquidity may not be able to reduce their private savings to offset the mandate's impact. In this section, I assess the savings effects of the mandate on workers who are subject to such liquidity constraints.

I explore the role of liquidity by allowing for heterogeneous responses across workers with different amounts of financial assets or household income.[24] The idea

[22]Note that the tax data extend back to 2002, but I exclude that year from the analysis because private savings cannot be calculated without observing wealth in the previous year.

[23]As the 2005 reform lowered the threshold by around CHF 6,000, this reduced the contribution cap just below the cutoff (20% of net earnings) by more than CHF 1,000.

[24]Household income includes all types of income that an individual or married couple receive in a given year. Besides their own earnings, the spouse's earnings from employment are by far the most important income source for workers in the estimation sample. 67% of them are married, with partners receiving CHF 79,500 in employment income, on average. Averaging across all workers, with

is that drawing down liquid assets and resource sharing within the household can enable workers to hold consumption constant despite larger pension contributions. To implement this, I split up the estimation sample by terciles of financial wealth in the previous year (as current wealth may be affected by treatment status) and household income, respectively, and re-estimate the effect of the mandate on total savings, overall private savings, and overall voluntary retirement savings separately for these subsamples.[25] Reassuringly, I obtain similar results when using financial assets or household income as a proxy for liquidity constraints.

To the extent that saving decisions are made jointly within the household, heterogeneity by household income also sheds light on how saving responses vary across the income distribution. This allows me to investigate whether the effects estimated for workers with earnings close to the mandate threshold extrapolate to higher-income individuals.

Total Savings. Panel A of Figure 6 plots the estimated effects on total savings for subsamples of workers who differ in their likelihood to be affected by liquidity constraints. While splitting the sample reduces precision, the pattern of point estimates is consistent with the notion that liquidity-constrained workers cannot fully offset mandatory retirement contributions by reducing other types of savings. For workers in the bottom tercile, I estimate that the mandate increases total savings by CHF 210 when using financial wealth and by CHF 340 when using household income as a proxy. By contrast, the point estimate for total savings is close to zero for workers in the top tercile, implying full crowding-out in the absence of liquidity constraints.

Contrary to the limited average effect, this provides evidence that the mandate increases total savings of workers who are not able to fully finance the forced pension contributions by drawing down liquid assets or receiving financial support from their partner. To further explore the variation in saving responses by degree of liquidity constraints, I now consider the effects on private savings and voluntary pension contributions for the same subsamples of workers.

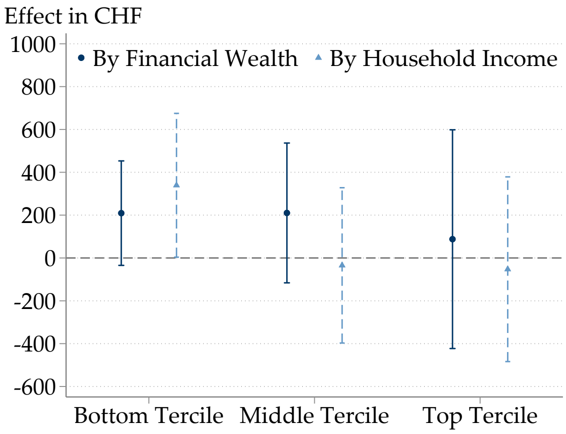
Private Savings. Panel B shows the estimated effects on private savings. In line with the results for total savings, I find that the drop in private savings is strongly attenuated and statistically insignificant for workers in the bottom tercile. The point estimate is around CHF –170 using financial wealth and CHF –30 using household income as a proxy. Again, this suggests that workers who are subject to liquidity constraints are less able to reduce their private savings to offset mandated pension contributions. For workers in the top tercile, I estimate a large and statistically significant

and without a partner, mean household income is CHF 79,800, of which 66% is accounted for by the partner's employment income and 27% by the worker's own employment income.

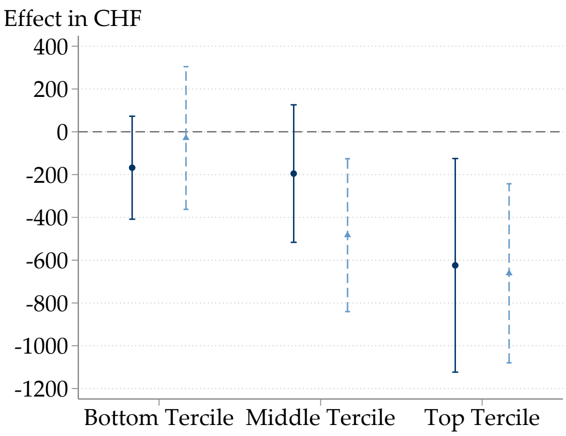
[25] The tercile cutoff values for financial wealth are CHF 4,700 and CHF 33,000. The tercile cutoffs for household income are CHF 46,100 and CHF 100,200.

Figure 6: Heterogeneity in Saving Responses by Liquidity Constraints

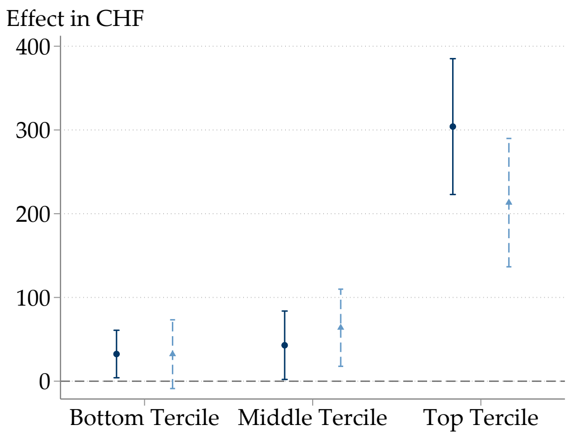
(a) total savings



(b) private savings



(c) voluntary pension savings



Notes: Coefficient plots showing the effects of the pension plan mandate by tercile of financial wealth in the previous year and household income, respectively, using data from 2005 to 2017. Overall sample includes 597,120 worker-year observations. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel, separately for the tercile groups. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

decline in private savings by more than CHF 600 using either proxy. Strikingly, the magnitude of this reduction exceeds the mandatory retirement contributions. Next, I assess whether this disproportionate drop in private savings is driven by top-tercile workers increasing their voluntary pension savings when facing the higher contribution cap.

Voluntary Retirement Savings. To examine how liquidity affects the extent to which workers above the threshold take advantage of the increased contribution cap for private pension savings, Panel C displays the estimated effects on voluntary retirement savings. Irrespective of whether financial wealth or household income is used as a proxy, the positive effect is highly concentrated among workers in the top tercile. This explains why the drop in private savings in the top tercile more than compensates for the mandatory retirement contributions: these workers are reallocating some of their private savings into preferentially taxed private pension accounts in response to the higher contribution cap. For workers further down the distribution of financial wealth or household income, I estimate much smaller increases in voluntary pension savings. Presumably, this is both due to liquidity constraints and because they face lower marginal income tax and wealth tax rates which reduces the incentive to make these contributions.

Summary and Discussion. I find evidence that liquidity-constrained workers are less able to reduce their private savings to offset the retirement contributions required by the mandate. As a consequence, the mandate increases their overall savings and, by implication, lowers their consumption spending by around CHF 200–350, corresponding to 1–1.75% of earnings. Workers with high levels of financial wealth or household income respond to the mandate (and the associated increase in the private pension contribution cap) by cutting private savings by a disproportionate amount in order to offset the mandated pension contributions and channel more savings into tax-advantaged private pension accounts, while holding overall savings and consumption constant. These results are consistent with Choukhmane and Palmer (2024) who find that workers with low initial deposit balances respond to increased default pension contributions mainly by cutting spending, whereas workers with high balances shift savings into pension accounts without reducing spending much.

The fact that workers with high household income completely offset mandatory pension contributions by reducing private savings suggests that crowd-out rates further up the earnings distribution are likely even larger than the 73% average crowd-out rate I estimate for workers near the threshold. This heterogeneity may explain why previous research on mandatory pension plans has found limited crowding-out among Danish workers with very low annual earnings of around $5,000 (Chetty et al., 2014) and stronger crowding-out among faculty at US universities (Card and Ransom, 2011; Friedberg, Leive and Cai, 2024).

7 Conclusion

In half of all OECD countries, workers are required to contribute to a pension plan. But there is limited evidence whether these mandates increase overall savings, mainly due to a lack of comprehensive data on savings. This paper presents novel evidence on saving responses to mandatory pension plans across the entire portfolio, drawing on tax microdata on wealth and savings from the Swiss canton of Bern and leveraging a discontinuity in mandatory pension contributions for identification.

I find that the mandate has limited effects on total savings, as workers offset mandatory retirement contributions by strongly reducing private savings. The drop in private savings is primarily driven by financial assets. I do not find an impact on savings in other types of assets such as business wealth, property, other wealth, and debt. I estimate a precise null effect on voluntary pension savings once the change in the contribution cap on preferentially taxed private pension savings is accounted for. Consistent with standard models of saving under borrowing constraints, liquidity-constrained workers reduce their private savings by less, and hence moderately increase their total savings in response to the mandate. Workers with high levels of financial wealth or household income cut their private savings disproportionately to offset the mandated retirement contributions and funnel more savings into tax-advantaged private pension accounts.

The strong substitution between mandatory pension contributions and private *non-retirement* savings highlights that it is crucial to assess the effects of pension policies throughout the entire portfolio. Prior research that focuses on the impact of mandatory pension plans on retirement savings due to data limitations therefore estimates a lower bound for the overall crowd-out rate (Card and Ransom, 2011; Friedberg, Leive and Cai, 2024). The result that liquidity constraints weaken the crowding-out response can potentially explain why previous work studying workers at different points of the income distribution has found a broad range of savings effects (Card and Ransom, 2011; Chetty et al., 2014; Friedberg, Leive and Cai, 2024). An important contextual feature of mandatory pension plans in Switzerland is that they are fully funded, with a strong and salient tax-benefit linkage. This likely contributes to their substitutability with private savings.

These findings have implications for pension policy. If mandatory pension plans are intended to raise overall savings levels, their effectiveness is limited. The exception are liquidity-constrained workers, but their savings gains come at a particularly high cost in terms of marginal utility of current consumption. Despite the modest impact on total savings, the mandate may still improve financial preparedness for retirement by shifting savings from financial assets to less liquid pension accounts. However, it is unclear whether this portfolio reallocation makes workers better off, as the lower share of liquid assets leaves them more exposed to negative financial shocks during their working lives (Choukhmane and Palmer, 2024). In the long run, potential differences in after-tax rates of return between the two types of assets may also play an important role. A thorough examination of these welfare considerations is beyond the scope of this paper but constitutes a fruitful avenue for future research.

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Online Appendix

Saving Responses to Mandatory Pension Plans[1]

David Burgherr[2]

July 2025

Contents

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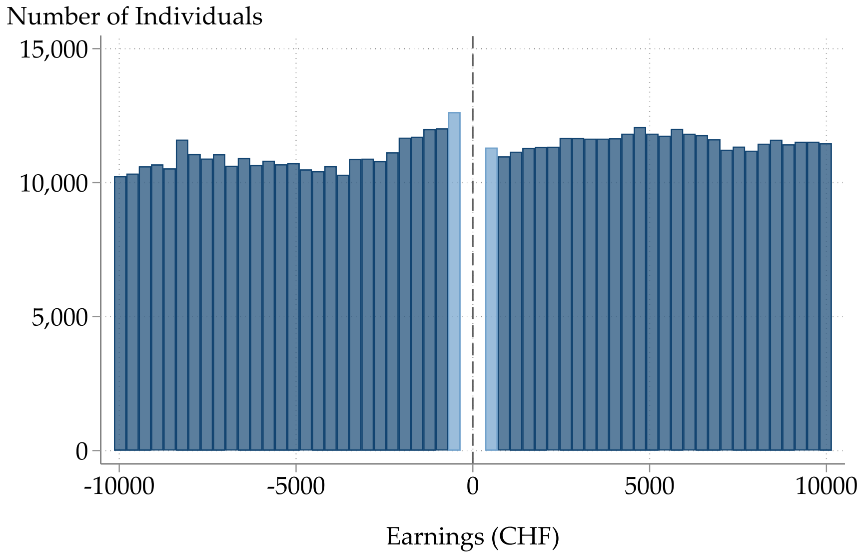
A Additional Figures and Tables

Figure A.1: Mandatory Pension Contributions as a Function of Earnings

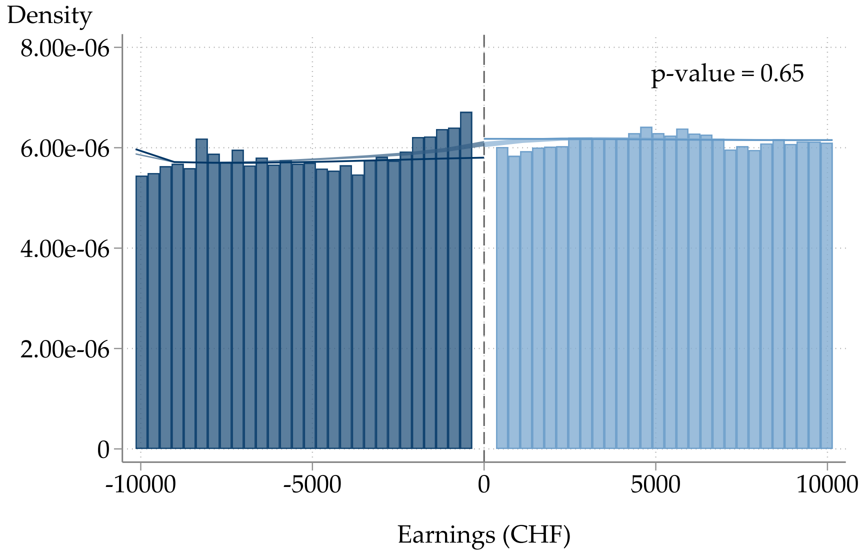
Notes: The figure displays mandated occupational pension contributions as a function of gross earnings for an employee between 45 and 54 years of age in 2017. Contributions are computed according to the statutory contribution schedule, i.e, applying a contribution rate of 15% to qualifying earnings. The relationship is very similar in other years, with slight variations depending on the year-specific parameters of the contribution schedule listed in Appendix Table B.1.
Source: Author's illustration based on information from the Federal Social Insurance Office.

Figure A.2: Density Test for Earnings in Main Job

(a) frequency distribution



(b) density test

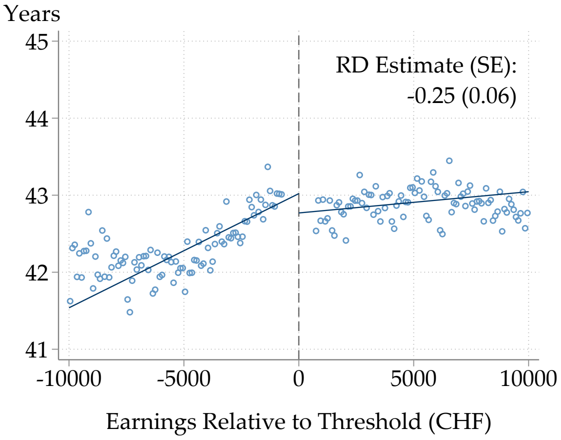


Notes: Panel A displays the frequency distribution of gross earnings in the main job (running variable) within CHF 10,000 of the threshold of the pension plan mandate using data from 2005 to 2017. Panel B plots the results of the density test proposed by Cattaneo, Jansson and Ma (2018, 2020) for the same observations. The bars represent the histogram of the distribution; the lines represent bias-corrected density estimates using local linear regression; the shaded areas represent valid confidence bands. The dashed vertical line indicates the mandate threshold. Workers within CHF 350 of the cutoff are excluded because their gross earnings cannot be imputed unambiguously. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions.

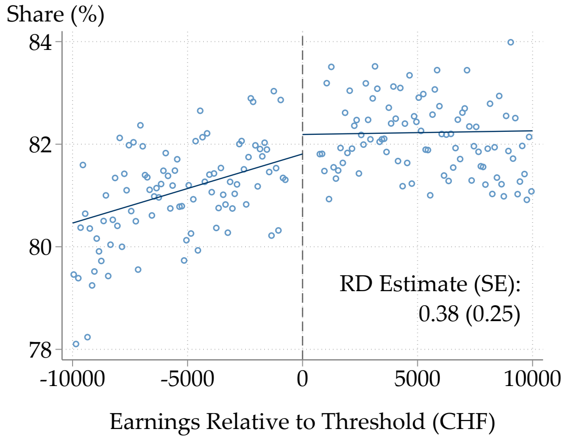
Source: Author's calculations based on administrative tax data from the canton of Bern.

Figure A.3: Continuity of Predetermined Characteristics

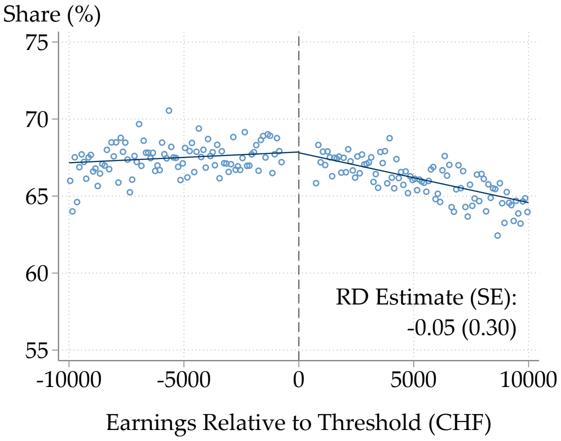
(a) Age



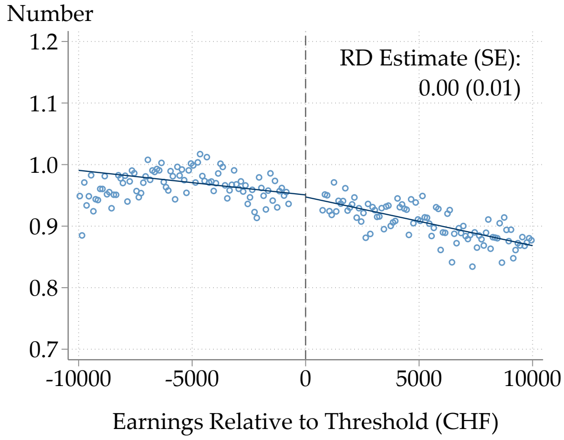
(b) Share of women



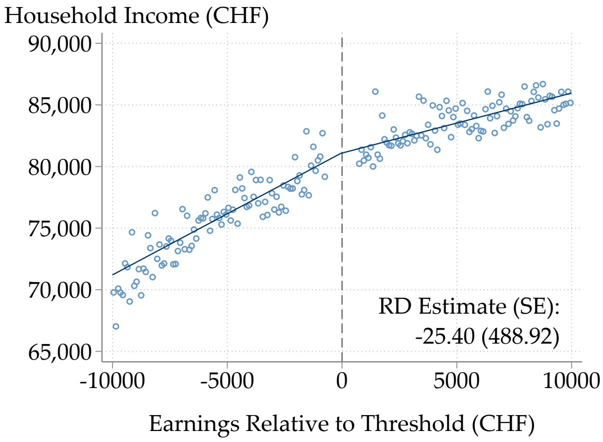
(c) Share married



(d) Number of children



(e) Household income

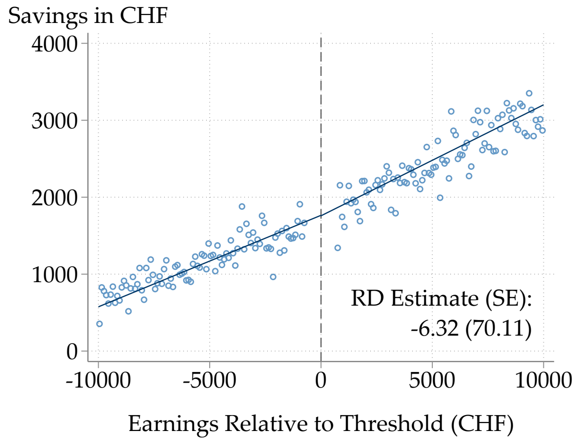


Notes: Regression discontinuity plots for a set of predetermined characteristics using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) without control variables, using a triangular kernel. The running variable is recentered around the mandate threshold. See Section 4.2 for more details on estimation.

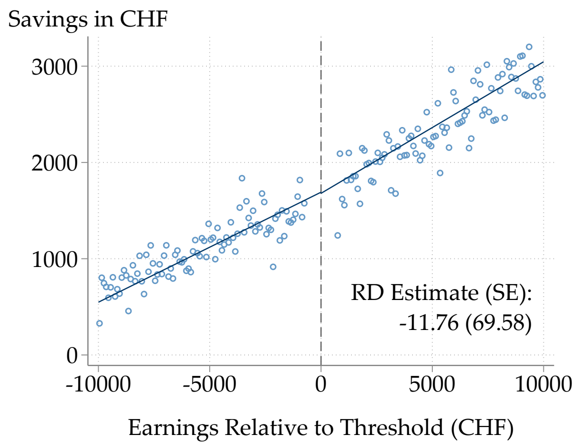
Source: Author's calculations based on administrative tax data from the canton of Bern.

Figure A.4: Placebo Test: Effects of Mandate on Workers Aged 20–24

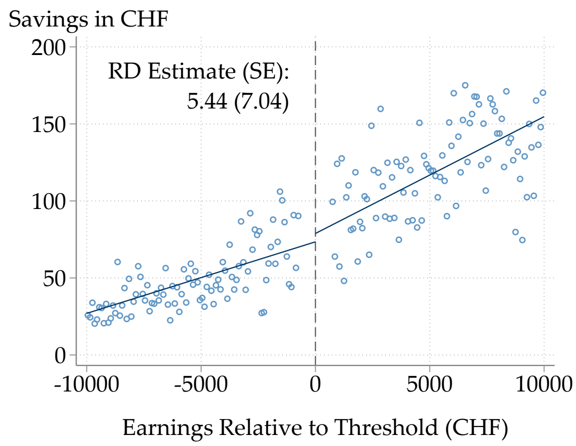
(a) total savings



(b) overall private savings



(c) overall voluntary pension savings

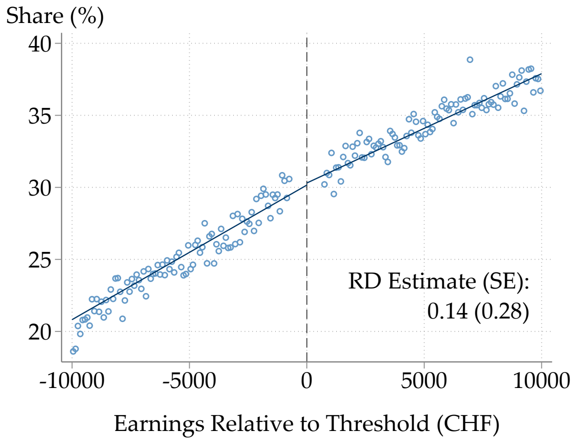


Notes: Regression discontinuity plots showing the effects of the pension plan mandate on a placebo sample of workers who are 20–24 years old and thus not affected by the mandate, using data from 2005 to 2017. Effective estimation sample includes 169,443 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on estimation.

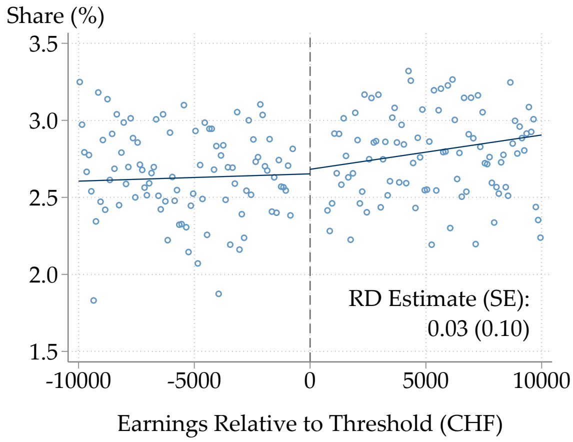
Source: Author's calculations based on administrative tax data from the canton of Bern.

Figure A.5: Effects of Mandate on Intervals of Private Pension Contributions

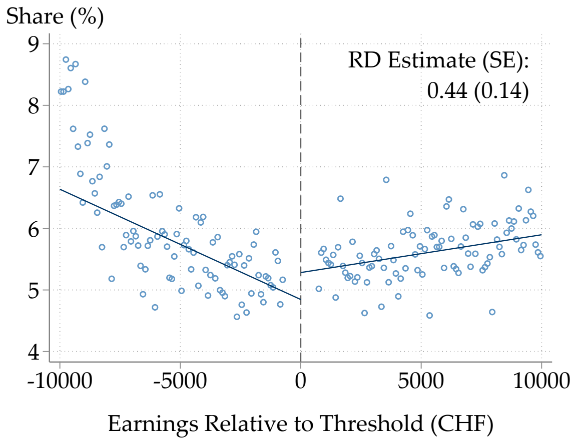
(a) contribution > 0



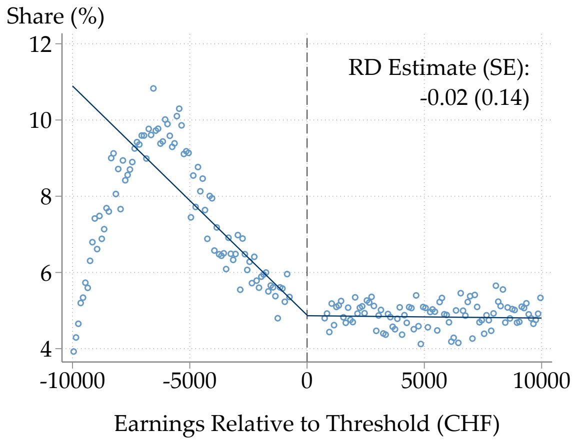
(b) contribution $\in (0, 1000]$



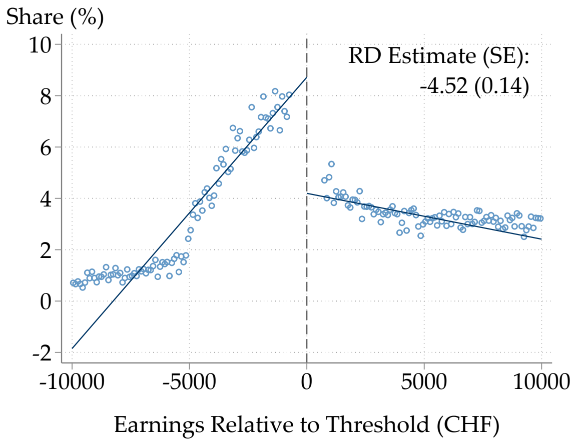
(c) contribution $\in (1000, 2000]$



(d) contribution $\in (2000, 3000]$



(e) contribution $\in (3000, 4000]$



(f) contribution $\in (4000, 5000]$

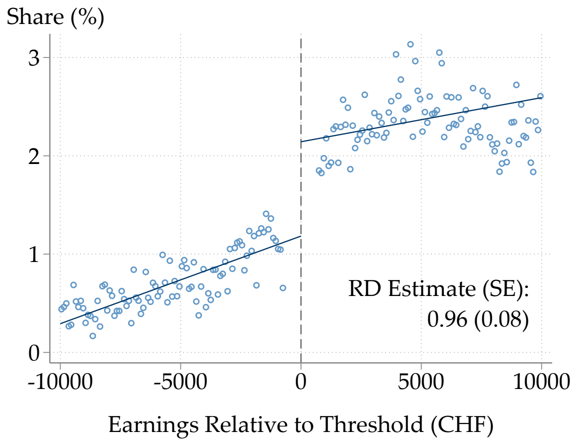
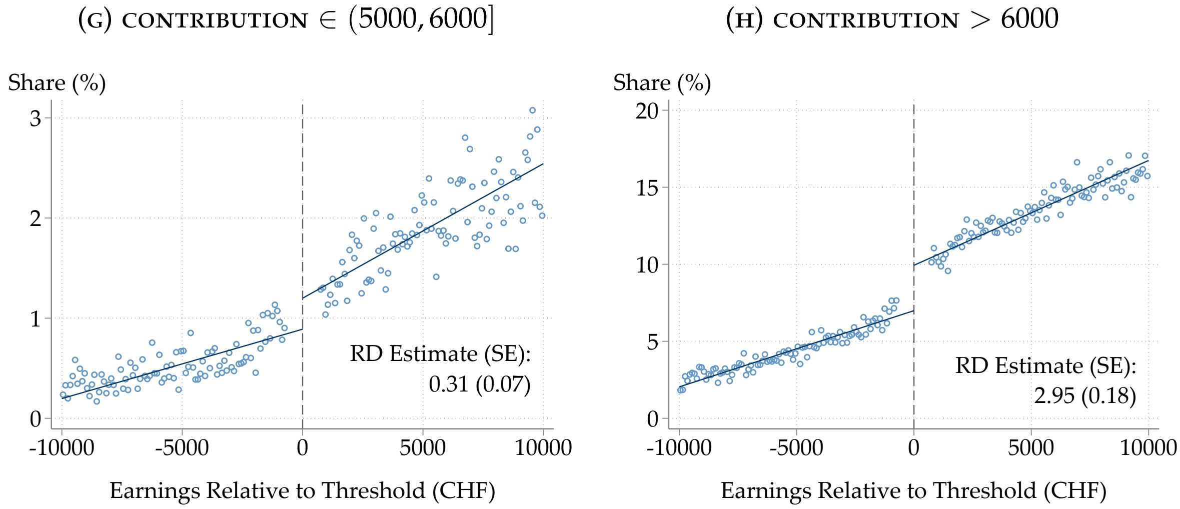


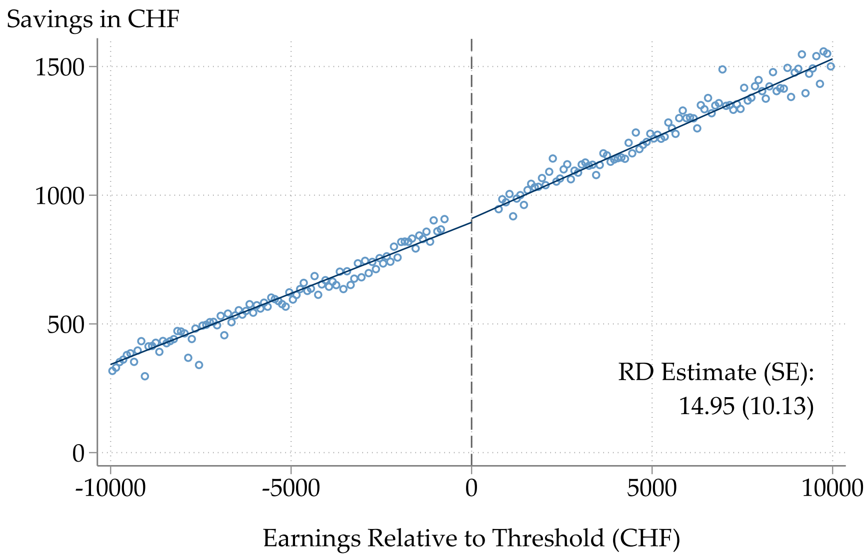
Figure A.5: Effects of Mandate on Intervals of Private Pension Contributions (cont.)



Notes: Regression discontinuity plots showing the effect of the pension plan mandate on the share making any private pension contributions (Panel A) and on the share making contributions in CHF 1,000 intervals (Panels B through H), using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

FIGURE A.6: EFFECT OF MANDATE ON CAPPED PRIVATE PENSION SAVINGS

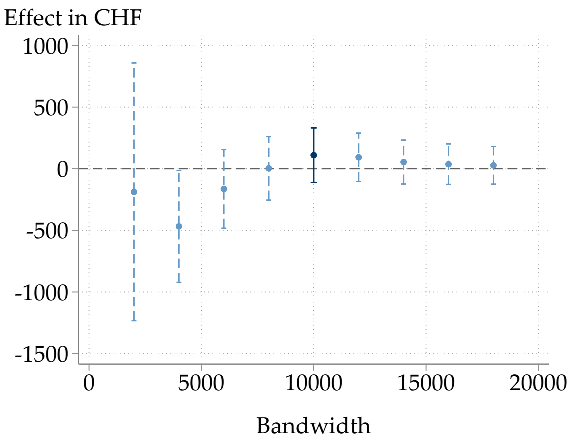


Notes: Regression discontinuity plot showing the effect of the pension plan mandate on private pension savings using data from 2005 to 2017, after imposing the same contribution cap rule below and above the mandate threshold. Effective estimation sample includes 597,120 worker-year observations. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

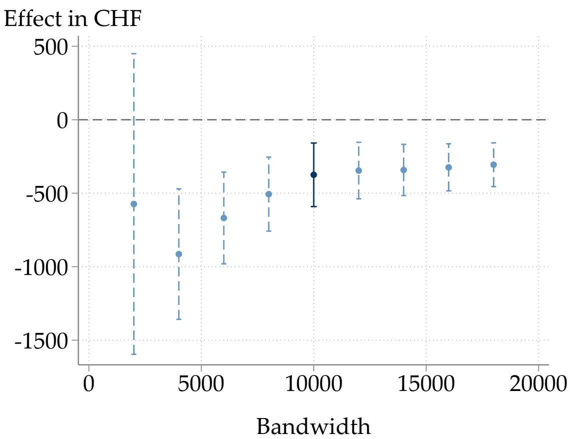
Source: Author's calculations based on administrative tax data from the canton of Bern.

Figure A.7: Sensitivity of RD Estimates to Bandwidth

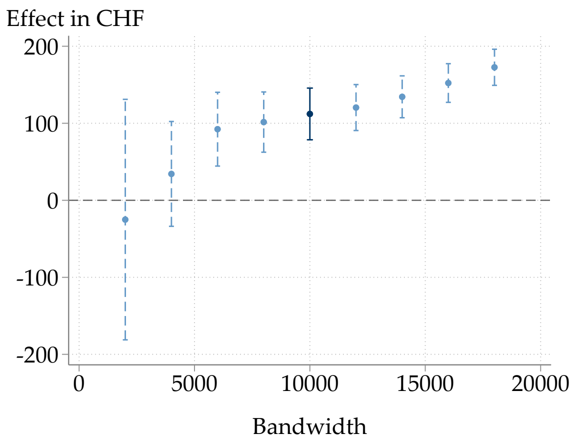
(a) total savings



(b) overall private savings



(c) overall voluntary pension savings

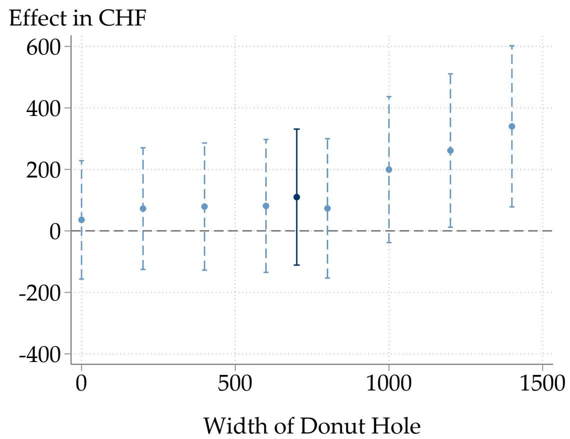


Notes: Coefficient plots showing the effects of the pension plan mandate depending on the bandwidth choice, using data from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel. The dark marker represents the main result using a bandwidth of CHF 10,000; the dashed light markers represent estimates using alternative bandwidths. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

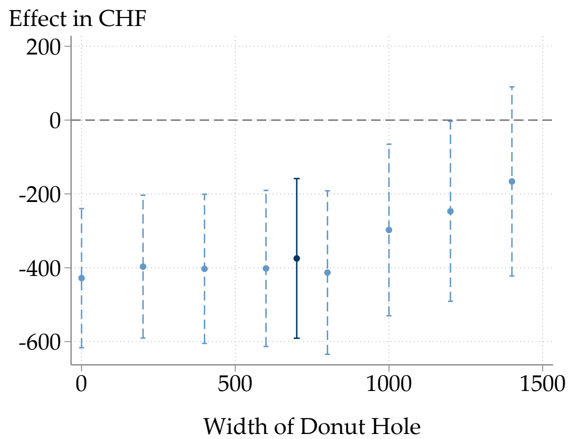
Source: Author's calculations based on administrative tax data from the canton of Bern.

Figure A.8: Sensitivity of RD Estimates to Size of Donut Hole

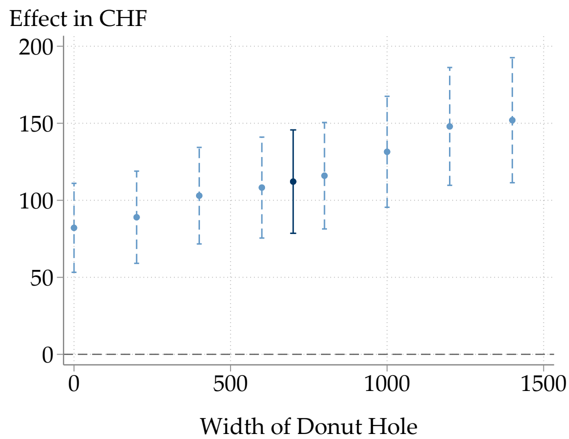
(a) total savings



(b) overall private savings



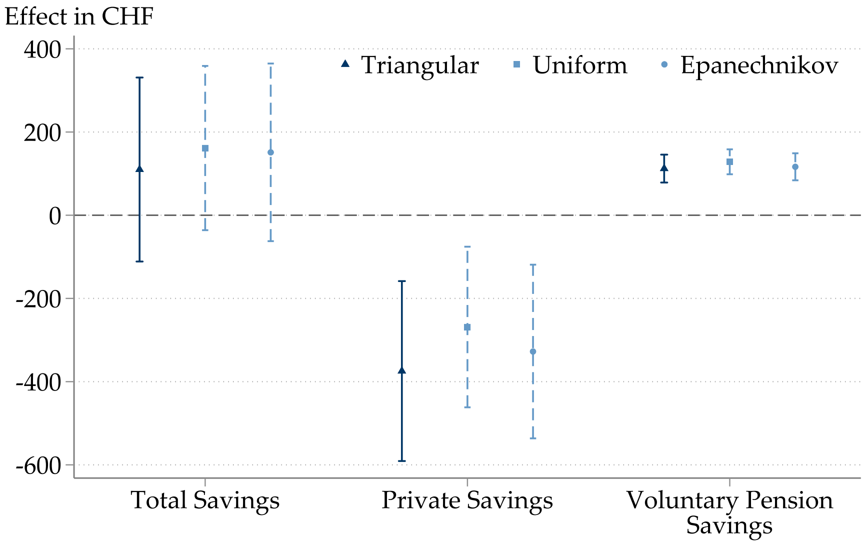
(c) overall voluntary pension savings



Notes: Coefficient plots showing the effects of the pension plan mandate depending on the size of the donut hole removed from the estimation sample, using data from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel. The dark marker represents the main result using a donut hole of CHF 700; the dashed light markers represent estimates using alternative donut hole sizes. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

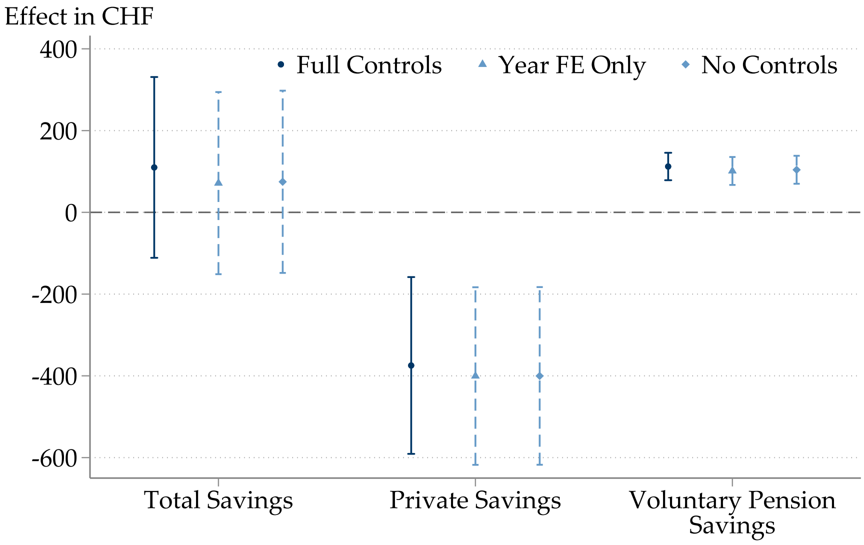
Figure A.9: Sensitivity of RD Estimates to Kernel



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the kernel weights, using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result using a triangular kernel; the dashed light markers represent estimates using alternative kernels. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

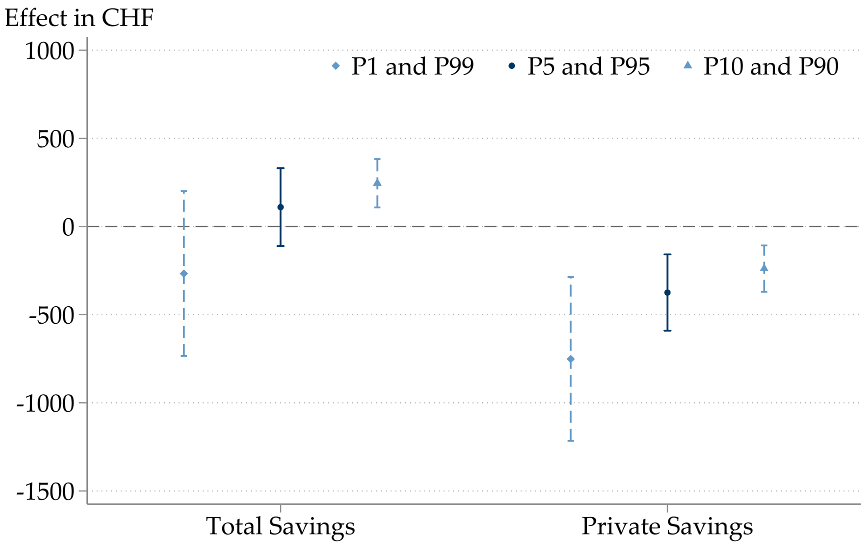
Figure A.10: Sensitivity of RD Estimates to Control Variables



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the set of controls included, using data from 2005 to 2017. Effective estimation sample includes 597,120 worker-year observations. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result estimated with the full set of controls; the dashed light markers represent estimates including only year fixed effects or no controls at all. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

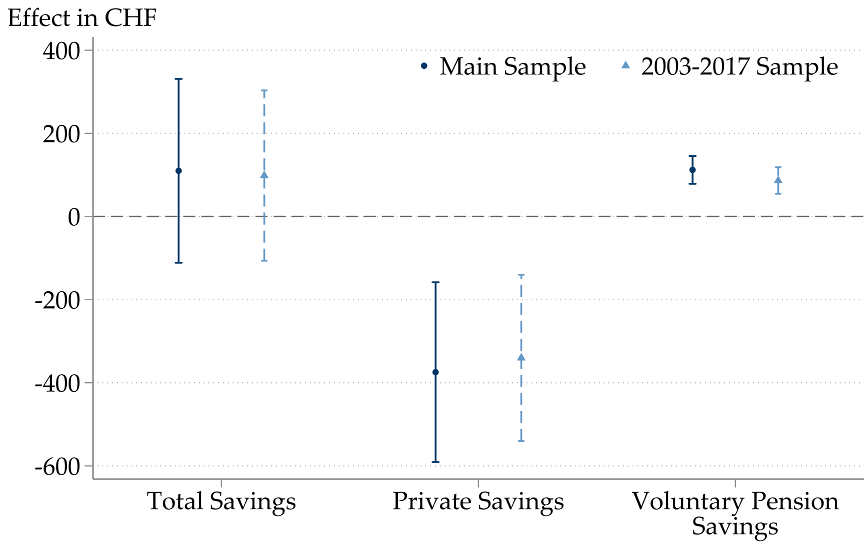
Figure A.11: Sensitivity of RD Estimates to Winsorization



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the winsorization cutoffs. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result using outcomes winsorized at percentiles 5 and 95; the dashed light markers represent estimates using outcomes winsorized at percentiles 1 and 99 or percentiles 10 and 90, respectively. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

Figure A.12: Sensitivity of RD Estimates to Sample Period



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the sample period. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result using data from years 2005–2017; the dashed light markers represent estimates using data from the full period 2003–2017. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

Table A.1: Summary Statistics

	Mean	Median	Std. Dev.
Panel A: Demographics			
Age	42.61	43	10.22
Female	0.82	1	0.39
Married	0.67	1	0.47
Number of Children	0.93	1	1.07
Panel B: Income			
Household Income	79,783	79,585	70,182
Total Individual Income	25,343	24,123	31,176
Gross Earnings Main Job	20,689	21,146	6,013
Gross Earnings Side Job	734	0	3,043
Self-Employment Income	684	0	5,498
Other Income	3,237	59	30,056
Panel C: Wealth (Stock Variables)			
Net Wealth	60,722	17,648	102,707
Financial Wealth	42,661	15,054	63,191
Business Wealth	488	0	1,902
Property Wealth	87,469	0	122,001
Other Wealth	2,312	0	4,525
Debt	-83,318	-2,000	116,742
Panel D: Savings (Annual Flows)			
Total Savings	3,966	1,114	17,532
Mandatory Pension Savings	245	232	276
Voluntary Pension Savings	1,203	0	2,821
Private Pension Savings	1,132	0	2,113
Occupational Pension Buy-ins	72	0	1,706
Private Savings	2,518	272	17,047
Financial Savings	1,589	0	11,451
Business Savings	66	0	1,389
Property Savings	2,903	0	14,168
Other Savings	18	0	1,953
Debt Savings	-610	0	5,653
Number of Observations		597,120	
Number of Individuals		191,400	

Notes: Mean, median, and standard deviation of demographic, income, wealth, and savings variables in the estimation sample, pooling data from 2005 to 2017. Estimation sample includes workers aged 25–60 with earnings within CHF 10,000 of the mandate threshold. All monetary values are reported in Swiss francs. All wealth variables and all private savings variables are winsorized at percentiles 5 and 95, except business savings and property savings which are winsorized at percentiles 2.5 and 97.5. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions.
Source: Author's calculations based on administrative tax data from the canton of Bern.

B The Swiss Old-Age Provision System

This appendix provides a comprehensive description of the Swiss old-age provision system, focusing on years 2002 to 2017 which is the period covered by the tax records.[3] Similar to other countries, the Swiss old-age provision system has three pillars: (i) a pay-as-you-go system ("old-age insurance"), (ii) an occupational pension system that is compulsory above an earnings threshold, and (iii) voluntary private pension accounts with a contribution cap. Appendix Table B.1 reports year-specific information on key parameters of the old-age provision system since the introduction of the occupational pension system in 1985.

B.1 Old-Age Insurance

Old-age insurance is organized as a pay-as-you-go scheme and compulsory for all individuals living or working in Switzerland between the age of 18 years and the statutory retirement age which is 64 years for women and 65 years for men. Between 2002 and 2017, the contribution rate applied to gross earnings was constant at 8.4%, of which employer and employees each pay half. The contribution of employees are deducted from their earnings by the employer and directly transferred to the social insurance administration. There is no cap on contributions, even if they do not lead to higher benefits. While benefit levels do depend on average earnings and the number of contribution years, the maximum benefit level is only double the minimum benefit level. From 2002 to 2017, the minimum annual benefit level was gradually increased from CHF 12,360 to CHF 14,100, while the maximum benefit level, accordingly, was rising from CHF 24,720 to CHF 28,200. This is relevant because key parameters of the occupational pension system are defined with respect to the maximum pension from old-age insurance as I explain in more detail in the next section. Benefit levels are usually adjusted every other year based on the evolution of an index reflecting the arithmetic mean of nominal wage growth and inflation.

B.2 Occupational Pension System

In contrast to old-age insurance, the occupational pension system is fully funded. Female employees between 25 and 64 years of age as well as male employees between 25 and 65 years of age must be enrolled in an occupational pension plan by their employer if the annual gross earnings in their main job with the same employer

[3] For more information on the Swiss old-age provision system and relevant changes over time, see the website of the Federal Social Insurance Office: https://www.bsv.admin.ch/bsv/en/home/social-insurance.html [accessed on 20 October 2021].

exceed the threshold specified in the law. Employees cannot opt out of contributing to an occupational pension account if they meet the legal criteria.

Occupational pension funds are a key tool for wealth accumulation in Switzerland. Savings in occupational pension accounts total around 8% of GDP per year. Wealth accumulated in those accounts reached almost CHF 900 billion or 130% of GDP in 2017 (Federal Statistical Office, 2019).[4] According to data published by the Swiss National Bank, insurance and pension schemes – which mainly consist of capital in occupational pension funds – accounted for about 23% of total household wealth in Switzerland in 2020 (Annaheim and Heim, 2021).

The mandate threshold determining enrollment into occupational pension plans provides the identifying variation for the empirical strategies employed in the paper. From 1985 until 2004, the threshold had been equal to the maximum pension from old-age insurance in order to avoid overinsurance of the salary already covered by old-age insurance. As part of a multi-step reform of the occupational pension system that was implemented between 2004 and 2006, the threshold was lowered to 3/4 of the maximum benefit level in 2005, resulting in a drop of the threshold from CHF 25,320 in 2004 to CHF 19,350 in 2005. Before and after the reform, the threshold was gradually increased in proportion to the rising maximum old-age insurance benefit level.[5] Note that the mandate only applies to earnings in the main job.[6] Employees with multiple jobs who exceed the mandate threshold only when summing up multiple salaries can join an occupational pension fund on a voluntary basis. Yet, the number of individuals making use of that option seems to be negligible (Ecoplan, 2010).[7] Self-employed individuals can also voluntarily join an occupational pension plan in which case they subject themselves to the cap on tax-deductible private pension savings that applies to employees (see Section B.3).

To compute mandatory occupational pension contributions, the qualifying earnings are multiplied by the age-specific contribution rate.[8] Qualifying earnings are equal to gross earnings minus a deduction. Similar to the mandate threshold, the deduction's purpose is to prevent overinsurance. It had been equal to the threshold between 1985 and 2004, but in the 2005 reform the threshold was reduced by more than the deduction. Since then, the deduction equals 7/8 of the maximum benefit

[4]Switzerland's GDP in 2017 was CHF 694 billion according to data from the State Secretariat for Economic Affairs: https://www.seco.admin.ch/seco/en/home/wirtschaftslage---wirtschaftspolitik/Wirtschaftslage/bip-quartalsschaetzungen-/daten.html [accessed on 19 October 2021].

[5]Figure B.1 displays the evolution of the earnings threshold since 1985.

[6]If employees are not working for the same employer for the whole year, the mandate threshold applies to the hypothetical earnings that they would have received if they had worked at that salary for the full year. However, employees must be on a permanent contract or work for the same employer for at least three months. I cannot distinguish these individuals from those who are not subject to the mandate because the tax data only contain information on annual earnings (see Appendix Section C.2).

[7]This finding holds up in more recent data, see Schöchli (2021).

[8]See Equation (4) in Appendix Section C.4 for the computation of occupational pension savings.

from old-age insurance while the threshold is 3/4, as mentioned earlier. The deduction ranges between CHF 22,575 and CHF 25,320 during the sample period.

The statutory contribution rates are 7% for employees aged 25–34, 10% for those aged 35–44, 15% for those aged 45–54, and 18% for women aged 55–64 and men aged 55–65.[9] Employers are legally obliged to pay at least 50% of the contribution. Occupational pension funds have some autonomy in designing their plans if they want to go beyond the minimum standards defined in the law (Dorn and Sousa-Poza, 2003).

For employees whose earnings are only slightly higher than the mandate threshold, there are minimum qualifying earnings equivalent to 1/8 of the maximum old-age insurance pension, varying between CHF 3,090 in 2002 and CHF 3,525 in 2017. Their contributions are calculated by applying the contribution rate to minimum qualifying earnings, not just to the marginal earnings above the threshold or deduction. Hence, there is a discontinuity in occupational pension savings at the cutoff.

If employees' earnings exceed an upper bound, there are maximum qualifying earnings equal to the upper bound minus the deduction. The upper bound is defined as three times the maximum pension from old-age insurance, gradually increasing from CHF 74,160 in 2002 to CHF 84,600 in 2017.[10] Furthermore, many occupational pension plans allow one-off "buy-ins" that individuals can voluntarily make beyond the mandatory contributions. These can also be deducted from taxable income.

To illustrate the calculation of contributions, Appendix Figure A.1 displays the relationship between mandatory occupational pension contributions and gross earnings for an employee aged 45–54 (contribution rate of 15%) in 2017. The discontinuity in mandatory pension savings at the cutoff is clearly visible. This relationship follows the same pattern in all other years with slight differences based on the year-specific parameters listed in Appendix Table B.1.

Retirees can choose to receive occupational pension benefits as some combination of a lifelong annuity and a lump-sum payment. By law, they are entitled to receive at least a quarter of their occupational pension capital as a lump sum.

The transformation of pension capital into an annuity is subject to a legally defined minimum conversion rate. The minimum conversion rate is only binding for the calculation of benefits based on pension capital deriving from qualifying earnings. It does not apply to voluntary occupational pension savings. The rate was reduced

[9] Women's legal retirement age in the occupational pension system was raised from 62 to 64 years in 2005 in order to be aligned with the rules of old-age insurance. Because of the lower retirement age, the statutory contribution rates applied to slightly different age groups for women before 2005 – specifically, 7% for the age group 25–31 years, 10% for the age group 32–41 years, 15% for the age group 42–51 years, and 18% for the age group 52–62 years.

[10] Most occupational pension funds provide insurance for the portion of the salary above the upper bound on a voluntary basis (Bütler, 2009). This practice is not relevant for employees in the earnings range studied in this paper.

gradually from 7.2% to 6.8% over the sample period as a consequence of the reform implemented between 2004 and 2006. The conversion rate translates the occupational pension capital into annual benefit entitlements. Accordingly, CHF 100,000 are converted into an annuity of CHF 6,800–7,200 depending on the year-specific conversion rate. Old-age insurance and occupational pension benefits are designed to jointly replace around 60% of pre-retirement earnings.

The tax treatment of occupational pension savings is quite favorable. Contributions and investment returns are exempt from income tax, and there is no capital gains tax in Switzerland. Moreover, pension capital is exempt from wealth tax. Benefits are subject to tax: while standard income tax is levied on annuities, lump-sum receipts are taxed at special, rather advantageous rates.

B.3 Private Pension Savings

Individuals can make voluntary contributions to designated private pension accounts. Because these savings can be deducted from taxable income, they are subject to a contribution cap. Employees enrolled in an occupational pension plan are allowed to make annual contributions to private pension accounts of up to 8% of the upper bound of qualifying earnings in the second pillar (equivalent to 24% of the maximum old-age insurance pension benefit). This cap has gradually increased from CHF 5,933 in 2002 to CHF 6,768 in 2017. Self-employed individuals who voluntarily join an occupational pension plan are subject to the same cap on private pension savings. Individuals who are not enrolled in an occupational pension plan, whether employed or self-employed, can contribute up to 20% of their net income but at most 40% of the upper bound of qualifying earnings in the occupational pension system (ranging between CHF 29,664 in 2002 and CHF 33,840 in 2017) to private pension accounts.

Private pension accounts are set up independently from occupational pension accounts, either at a bank or insurance company. The law allows contributions to standard savings accounts as well as investments into stocks and other securities. Use of this savings vehicle is fairly widespread: private pension capital in designated accounts totalled CHF 121 billion or 17.4% of GDP in 2017.[11]

Until people enter retirement, they have limited access to the accumulated capital in these designated private pension accounts. Once eligible, disbursement can be in the form of an annuity or lump sum.

The tax treatment of private pension savings is similar to occupational pension savings: contributions can be deducted from taxable income, capital is exempt from

[11] Statistics on the occupational pension and private pension system are available from the Federal Social Insurance Office: https://www.bsv.admin.ch/bsv/en/home/social-insurance/bv/statistik.html [accessed on 20 October 2021].

wealth tax, while annuities are subject to income tax and lump-sum payments are taxed at preferential rates at the time of receipt.

B.4 Reform of the Occupational Pension System in 2004–2006

The 2004–2006 reform was the first major reform of the occupational pension system since its introduction in 1985, and it remains the only one to date.[12] It was implemented in three steps between 2004 and 2006. Before the reform, from 1985 to 2004, both the earnings threshold and the deduction for computing mandatory occupational pension savings had been equal to the maximum pension in the pay-as-you-go system. Accordingly, they had been increasing one-for-one with the rising maximum benefit level during that period, reaching CHF 25,320 in 2004.

As part of the reform, both the threshold and the deduction were lowered in 2005. Appendix Figure B.1 plots the evolution of the threshold since introduction of the occupational pension system in 1985, demonstrating the significance of the 2005 reduction. The 2005 reform reduced the threshold to 3/4 of the maximum pay-as-you-go pension. As a consequence, it fell from CHF 25,320 in 2004 to CHF 19,350 in 2005. Around 140,000 employees in Switzerland who otherwise would not have been subject to the mandate became covered due to the reform, the majority of which were female, working part-time, and had low hourly wages below CHF 25 (Ecoplan, 2010). This represents an increase in the number of workers covered by roughly 4%. The purpose of the reduction of the threshold was to improve financial preparedness for retirement among low-income workers.

The deduction was lowered to 7/8 of the maximum pay-as-you-go pension, i.e., to CHF 22,575 in 2005. This increased mandatory occupational pension savings at almost all earnings levels above the threshold. Lowering the deduction aimed to offset some of the decline in pension benefit levels caused by the decrease in the conversion rate implemented in the same reform.

The reform included further policy changes such as lowering the statutory rate for converting pension account balances into annuities from 7.2% to 6.8% over a transition period of ten years, increasing women's retirement age in the occupational pension system to 64 years in line with the rules in the pay-as-you-go system, and aligning the contribution rate schedule for women with that of men.

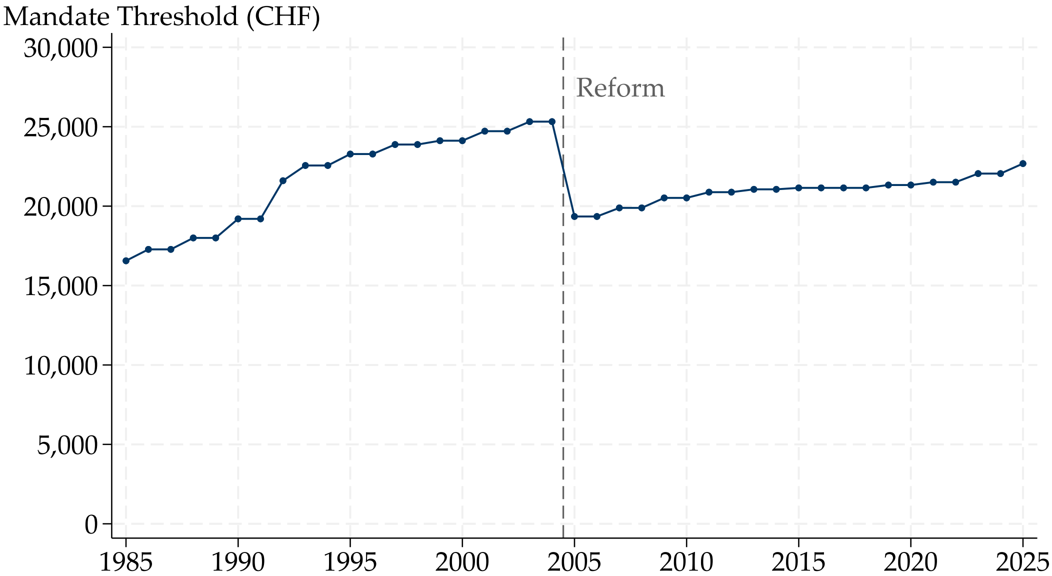
[12] Detailed information on the reform ("1. BVG-Revision") is available from the Federal Social Insurance Office: https://www.bsv.admin.ch/bsv/de/home/sozialversicherungen/bv/reformen-und-revisionen/revision-1-bvg.html [accessed on 30 October 2021].

Table B.1: Key Parameters of the Old-Age Provision System in Switzerland

	Old-age insurance		Occupational pension system				Private pension savings
Year	Minimum benefit	Maximum benefit	Threshold	Deduction	Upper bound	Min. qual. earnings	Contribution cap
1985	8,280	16,560	16,560	16,560	49,680	2,070	-
1986	8,640	17,280	17,280	17,280	51,840	2,160	-
1987	8,640	17,280	17,280	17,280	51,840	2,160	4,147
1988	9,000	18,000	18,000	18,000	54,000	2,250	4,320
1989	9,000	18,000	18,000	18,000	54,000	2,250	4,320
1990	9,600	19,200	19,200	19,200	57,600	2,400	4,608
1991	9,600	19,200	19,200	19,200	57,600	2,400	4,608
1992	10,800	21,600	21,600	21,600	64,800	2,700	5,184
1993	11,280	22,560	22,560	22,560	67,680	2,820	5,414
1994	11,280	22,560	22,560	22,560	67,680	2,820	5,414
1995	11,640	23,280	23,280	23,280	69,840	2,910	5,587
1996	11,640	23,280	23,280	23,280	69,840	2,910	5,587
1997	11,940	23,880	23,880	23,880	71,640	2,985	5,731
1998	11,940	23,880	23,880	23,880	71,640	2,985	5,731
1999	12,060	24,120	24,120	24,120	72,360	3,015	5,789
2000	12,060	24,120	24,120	24,120	72,360	3,015	5,789
2001	12,360	24,720	24,720	24,720	74,160	3,090	5,933
2002	12,360	24,720	24,720	24,720	74,160	3,090	5,933
2003	12,660	25,320	25,320	25,320	75,960	3,165	6,077
2004	12,660	25,320	25,320	25,320	75,960	3,165	6,077
2005	12,900	25,800	19,350	22,575	77,400	3,225	6,192
2006	12,900	25,800	19,350	22,575	77,400	3,225	6,192
2007	13,260	26,520	19,890	23,205	79,560	3,315	6,365
2008	13,260	26,520	19,890	23,205	79,560	3,315	6,365
2009	13,680	27,360	20,520	23,940	82,080	3,420	6,566
2010	13,680	27,360	20,520	23,940	82,080	3,420	6,566
2011	13,920	27,840	20,880	24,360	83,520	3,480	6,682
2012	13,920	27,840	20,880	24,360	83,520	3,480	6,682
2013	14,040	28,080	21,060	24,570	84,240	3,510	6,739
2014	14,040	28,080	21,060	24,570	84,240	3,510	6,739
2015	14,100	28,200	21,150	24,675	84,600	3,525	6,768
2016	14,100	28,200	21,150	24,675	84,600	3,525	6,768
2017	14,100	28,200	21,150	24,675	84,600	3,525	6,768
2018	14,100	28,200	21,150	24,675	84,600	3,525	6,768
2019	14,220	28,440	21,330	24,885	85,320	3,555	6,826
2020	14,220	28,440	21,330	24,885	85,320	3,555	6,826
2021	14,340	28,680	21,510	25,095	86,040	3,585	6,883

Notes: Year-specific parameters of the old-age provision system in Switzerland since the introduction of the occupational pension system in 1985. All values are reported in Swiss francs. The option to contribute to preferentially taxed private pension accounts was established in 1987. The cap on these private pension contributions applies to individuals enrolled in an occupational pension plan which includes all employees with earnings above the mandate threshold.
Source: Official information from the Federal Social Insurance Office.

Figure B.1: Earnings Threshold of the Occupational Pension Plan Mandate



Notes: The figure displays the evolution of the earnings threshold determining coverage of the occupational pension savings mandate from the introduction of the occupational pension system in 1985 to 2025. The dashed vertical line indicates the timing of the reform that lowered the threshold in 2005. Employees with gross earnings in their main job exceeding the threshold are subject to the savings mandate.

Source: Author's illustration based on information from the Federal Social Insurance Office.

C Data Appendix

In this appendix, I describe the administrative data provided by the tax authority of the canton of Bern in more detail. I start by discussing what variables are available at the individual or the household level, and how I split up couples into individual observations. Subsequently, I turn to the income, wealth, and savings data used in the paper. Finally, I discuss data cleaning and sample restrictions.

C.1 Individual-Level vs. Household-Level Information

In Switzerland, the relevant tax unit is the household. Thus, married couples jointly file one tax return. Because the savings mandate applies at the individual level, I follow the approach of Fagereng et al. (2020) and split up married couples into individual observations, equally assigning half of the value of income and wealth variables that are only reported at the household level to each partner. All wealth information is reported at the household level. Income variables reported at the household level include self-employment income, business income, financial income, and property income. Most important for my analysis, earnings from employment are reported at the individual level, as are transfer income and pension income. The same is true for occupational pension buy-ins and private pension savings.

C.2 Income

The pension plan mandate applies to gross earnings in the main job, before any deductions are applied. The tax records contain separate information for earnings in the main job and other jobs. The definition of earnings in the main job used by the tax administration overlaps to a large degree with the definition relevant for the mandate (see Appendix Section B.2). Both refer to total earnings with the same employer. There is a difference in that the tax authority includes positions at multiple employers as part of the main job if they are similar in terms of working hours or income received, while the mandate applies only to the main job at one employer. Yet, this distinction is unlikely to be relevant for many employees in the data, so the resulting measurement error should be limited.

Employees report earnings net of social insurance and occupational pension contributions to the tax administration because these are deducted directly from their salary by the employer. Therefore, I impute gross earnings based on net earnings recorded in the tax data and the year-specific social insurance and occupational pension schedules. Because social insurance and occupational pension contributions vary along the earnings distribution, the calculation must differentiate between certain earnings ranges.[13] Taking as an example an individual subject to the pension plan mandate for whom neither the minimum nor maximum qualifying earnings of the occupational pension system are binding, the calculation must take into account that social insurance contribution rates are applied to total gross earnings and occupational pension contribution rates are applied to gross earnings above the deduction. Overall, gross earnings in the main job, G_{it}, of employee i in year t can be imputed from net earnings as

$$G_{it} = \begin{cases} \frac{N_{it}}{1-i^e_{it}} & \text{if} \quad N_{it} < (C_t \times (1-i^e_{it}) - p^e_{it} \times M_t) \\ \frac{N_{it}}{1-i^e_{it}} \text{ or } \frac{N_{it}+p^e_{it}\times M_t}{1-i^e_{it}} & \text{if} \quad \begin{array}{l} N_{it} \geq (C_t \times (1-i^e_{it}) - p^e_{it} \times M_t) \text{ and} \\ N_{it} < (C_t \times (1-i^e_{it})) \end{array} \\ \frac{N_{it}+p^e_{it}\times M_t}{1-i^e_{it}} & \text{if} \quad \begin{array}{l} N_{it} \geq (C_t \times (1-i^e_{it})) \text{ and} \\ N_{it} < ((D_t + M_t) \times (1-i^e_{it}) - p^e_{it} \times M_t) \end{array} \\ \frac{N_{it}-p^e_{it}\times D_t}{1-i^e_{it}-p^e_{it}} & \text{if} \quad \begin{array}{l} N_{it} \geq ((D_t + M_t) \times (1-i^e_{it}) - p^e_{it} \times M_t) \text{ and} \\ N_{it} < (B_t \times (1-i^e_{it}) - p^e_{it} \times (B_t - D_t)) \end{array} \\ \frac{N_{it}+p^e_{it}\times(B_t-D_t)}{1-i^e_{it}} & \text{if} \quad N_{it} \geq (B_t \times (1-i^e_{it}) - p^e_{it} \times (B_t - D_t)), \end{cases} \tag{3}$$

where N_{it} denotes net earnings, C_t is the cutoff value of the pension plan mandate, D_t is the deduction, M_t are the minimum qualifying earnings, and B_t is the upper bound in the occupational pension system, while i^e_{it} represents the employee share of the social insurance contribution rate, and p^e_{it} represents the employee share of the age-specific occupational pension contribution rate.[14] Employees and employers each pay half of total social insurance contributions. Equally, I set the employee share of occupational pension contributions to 50 % which is the maximum employee share defined in the law. Some employers may bear more than 50% (which cannot be observed in the tax data), so this may result in slight measurement error.

Note that due to the discontinuity in mandatory occupational pension contributions, gross earnings slightly below and slightly above the mandate threshold can result in the same net earnings recorded in the tax data. Thus, gross earnings cannot be unambiguously imputed from net earnings for a small number of individuals (captured by the second line in Equation (3)). This problem only concerns a narrow

[13] Note that the definition of the different earnings ranges with respect to net earnings in Equation (3) is equivalent to the definition in terms of gross earnings in Equation (4).

[14] Social insurance contributions include contributions for old-age insurance, disability insurance, loss of earnings compensation, and unemployment insurance. Above a high earnings cutoff, unemployment insurance contribution rates are reduced in most years that I have data on. I account for the reduced unemployment insurance contributions when calculating gross earnings but omit it from Equation (3) because it does not matter for individuals in the earnings range of interest for this paper. More information on the social insurance system in Switzerland and historical contribution schedules are available on the website of the Federal Social Insurance Office: https://www.bsv.admin.ch/bsv/de/home/sozialversicherungen/ueberblick/beitraege.html [accessed on 23 October 2021].

earnings range of less than CHF 350 below and above the threshold (with the exact width depending on the age-specific contribution rate and the year-specific minimum qualifying earnings). I remove these observations from the sample used throughout the paper. Only in the robustness checks setting the donut hole to less than CHF 350 (Appendix Figure A.8), I include these observations, assigning them with equal probability the appropriate gross earnings below or above the mandate threshold.

The tax records also include information on other types of income which are less relevant for this paper such as self-employment income, business income, financial income, property income, transfer income, and pension income.

C.3 Wealth

The tax records contain detailed information on wealth by asset type, including business wealth, financial wealth, property wealth, other types of wealth, and debt. The Swiss wealth tax is quite comprehensive, covering all types of assets except for pension wealth in occupational and private pension accounts which is thus missing from the data. This is not a problem for the empirical analysis in this paper, because pension savings can be observed or imputed even without directly observing pension wealth. Information on pension contributions is sufficient because pension wealth generally cannot be accessed during the work life.

It needs to be noted that the valuation of property for tax purposes systematically underestimates the true market value. As a rule of thumb, residential property is valued at around 60% of its market value in Switzerland (OECD, 2018*c*), although in individual cases the valuation may deviate significantly from that benchmark. To analyze the discrepancy between tax value and market value of residential property, the tax administration of the canton of Bern conducted an analysis comparing the observed price of all housing transactions in a given year to the value of these properties in the tax records.[15] On average, the tax value of property was 71% of its market value in 2002. Because there was no revaluation during the sample period and property prices in Bern have generally been increasing, the tax value has gradually declined to about 55% of the market value in 2017.

C.4 Savings

Various types of pension and private savings could be affected by the occupational pension plan mandate. Some of these are directly observable in the tax data; others

[15]The analysis of property valuation for tax purposes is available on the website of the tax authority of the canton of Bern: https://www.sv.fin.be.ch/sv_fin/de/index/navi/index/steuersituationen/kauf-verkauf_liegenschaft/amtlicher_wert/allgemeine-neubewertung20.html [accessed on 23 October 2021].

need to be imputed based on information available in the dataset.

As mentioned earlier, mandatory occupational pension contributions are withheld at source by the employer, so they are not recorded in the tax data. I impute these by applying the statutory contribution rates – the sum of the employer and employee shares – to individuals' gross earnings in the main job. Based on the contribution schedule explained in Section 2.1, mandatory occupational pension savings, S_{it}^{occ}, of employee i in year t are calculated as

$$S_{it}^{occ} = \begin{cases} 0 & \text{if} \quad G_{it} < C_t \\ p_{it} \times M_t & \text{if} \quad G_{it} \geq C_t \quad \text{and} \quad (G_{it} - D_t) < M_t \\ p_{it} \times (G_{it} - D_t) & \text{if} \quad G_{it} \geq C_t \quad \text{and} \quad (G_{it} - D_t) \geq M_t \\ p_{it} \times (B_t - D_t) & \text{if} \quad G_{it} > B_t, \end{cases} \tag{4}$$

where p_{it} is the age-specific total occupational pension contribution rate, i.e., employer and employee share. All other variables are defined as in Equation (3). Appendix Figure A.1 illustrates how mandatory occupational pension savings are computed from gross earnings.

Private pension savings in preferentially taxed accounts as well as voluntary occupational pension buy-ins are directly observed at the individual level in the tax data, because they must be reported in the tax return in order to be deductible from taxable income.

I compute overall private savings as the change in net wealth relative to the previous year. Hence, private savings, $S_{i,t}^{priv}$, of individual i in year t are given by

$$S_{i,t}^{priv} = W_{i,t} - W_{i,t-1}, \tag{5}$$

where $W_{i,t}$ denotes net wealth of individual i in year t. Net wealth represents the difference between gross wealth and debt, and can be directly observed in the tax records. This savings measure corresponds to *gross savings* as it includes accrued capital gains from changes in asset prices (which are sometimes called "passive savings" in the literature) (see Fagereng et al., 2021). Note that my measure of private savings does not capture accrued gains on property because there has not been a revaluation of property during the observed period. I cannot separate gross savings into *net savings* (sometimes called "active savings") and capital gains because realized gains are not observed as Switzerland does not have a capital gains tax and I do not have data down to the level of single assets or transactions.[16] An implication of this sav-

[16]The latter is a common challenge in the literature on the measurement of savings. A few papers leverage very detailed administrative data to distinguish net savings and capital gains, including Bach, Calvet and Sodini (2020), Fagereng et al. (2020), and Fagereng et al. (2021). Fagereng et al. (2021) provide a helpful discussion of the distinction between gross and net savings and its implications, using both economic theory and an empirical application drawing on Norwegian data.

ings measure is that private savings are highly variable due to fluctuations in asset prices. Another reason for the variability of private savings is the purchasing timing of durable goods, services, and lumpy non-durables (Chetty et al., 2014).

Total savings are defined as the sum of all savings variables discussed above, including mandatory occupational pension savings, occupational pension buy-ins, private pension savings, and private savings.

C.5 Data Cleaning and Sample Restrictions

To prepare the data for the empirical analysis, I remove observations that are unreliable or not comparable to standard taxpayers (similar to Brunner, Meier and Näf, 2020). This group includes individuals who are only taxed for part of the year because they move abroad or arrive from abroad (1.9% of all observations), individuals who fail to hand in a tax return and are assessed by the tax authority (2.8%), duplicate observations for individuals in the same year (0.6%), and observations with obvious errors in the reported information (0.1%).